Singular Homology and Cohomology

Labix

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1 The Categorical Viewpoint

Recall that the category of topological spaces **Top** is complete and cocomplete. This means that all kinds of limits and colimits exists in **Top**. We have already seen the product space and disjoint union with their universal property as a limit / colimit. There are also more constructs that can be recognized / defined in terms of the universal property.

1.1 Categories of Spaces

Definition 1.1.1 (The Category of Pointed Topological Spaces) Define the category of pointed topological spaces $\mathbf{Top}_*$ to consist of the following data.

- The objects are a pair (X, x_0) where X is a topological space and $x_0 \in X$ is a chosen base point.
- For (X, x_0) and (Y, y_0) two pointed spaces, the morphisms

$$\text{Hom}_{\mathbf{Top}_*}((X, x_0), (Y, y_0)) = \{f : X \to Y \mid f \text{ is continuous and } f(x_0) = y_0\}$$

  are the continuous maps from X to Y such that base points are preserved.
- Composition is defined as the composition of functions such that base point is preserved.

Proposition 1.1.2 Let (X, x_0) and (Y, y_0) be pointed spaces. Then the product and coproduct of the two spaces in $\mathbf{Top}_*$ are

$$(X \times Y, (x_0, y_0)) \quad \text{and} \quad (X \vee Y, x_0 = y_0)$$

respectively.

Definition 1.1.3 (The Category of Pairs of Spaces) Define the category of pairs of topological spaces $\mathbf{Top}^2$ to consist of the following data.

- The objects are a pair (X, A) where X is a topological space $A \subseteq X$ is a subspace of X.
- For (X, A) and (Y, B) two pointed spaces, the morphisms

$$\text{Hom}_{\mathbf{Top}^2}((X, A), (Y, B)) = \{f : X \to Y \mid f \text{ is continuous and } f(A) \subseteq B\}$$

  are the continuous maps from X to Y such that subspaces are mapped to subspaces.
- Composition is defined as the composition of functions such that subspaces are mapped to subspaces.

Definition 1.1.4 (The Set of Homotopy Classes of Maps)
Let X, Y be spaces. Define the set of homotopy classes of maps from X to Y to be the quotient

$$[X, Y] = \frac{\{f : X \to Y \mid f \text{ is continuous }\}}{\sim}$$

where $f \sim g$ if f and g are homotopic.

Definition 1.1.5 (The Homotopy Category of Spaces)
Define the homotopy category of topological spaces **hTop** to consist of the following data.

- The objects are topological spaces.
- For X and Y two spaces, the morphisms

$$\text{Hom}_{\mathbf{hTop}}(X, Y) = [X, Y]$$

  are the homotopy classes of continuous maps from X to Y.
- Composition is defined by $[g] \circ [f] = [g \circ f]$ for $[f] \in \text{Hom}_{\mathbf{hTop}}(X, Y)$ and $[g] \in \text{Hom}_{\mathbf{hTop}}(Y, Z)$.

Definition 1.1.6 (The Set of Pointed Homotopy Classes of Maps)
Let $(X, x_0), (Y, y_0)$ be pointed spaces. Define the set of pointed homotopy classes of maps from (X, x_0) to (Y, y_0) to be the quotient

$$[(X, x_0), (Y, y_0)]_* = \frac{\{f : X \to Y \mid f \text{ is continuous and } f(x_0) = y_0\}}{\sim}$$

where $f \sim g$ if f and g are homotopic relative to x_0.

Definition 1.1.7 (The Homotopy Category of Pointed Spaces)
Define the homotopy category of pointed spaces $\mathbf{hTop}_*$ to consist of the following data.

- The objects are pointed topological spaces.
- For two pointed spaces (X, x_0) and (Y, y_0), the morphisms

$$\text{Hom}_{\mathbf{hTop}_*}((X, x_0), (Y, y_0)) = [(X, x_0), (Y, y_0)]_*$$

  are the pointed homotopy classes of continuous maps from (X, x_0) to (Y, y_0).
- Composition is defined by $[g] \circ [f] = [g \circ f]$ for $[f] \in \text{Hom}_{\mathbf{hTop}_*}((X, x_0), (Y, y_0))$ and $[g] \in \text{Hom}_{\mathbf{hTop}_*}((Y, y_0), (Z, z_0))$.

1.2 Categorical Constructs in the Category of Spaces

We have seen pushouts and pullbacks in Category Theory 1. Let us first recall them.

Definition 1.2.1 (Adjunction Spaces) Let X, Y be spaces and $A \subseteq X$ a subspace. Let $f : A \to Y$ be a map. Define the adjunction space of X and Y to be the space

$$X \amalg_f Y = \frac{X \amalg Y}{a \sim f(a)}$$

together with the quotient topology.

The prototypical example of adjunction spaces come from CW complexes. It is easy to notice that the data of obtaining the $(n+1)$-skeleton inductively is the same content of a pushout diagram:

$$\begin{tikzcd} S_\alpha^{n-1} \arrow[r, "\varphi_\alpha"] \arrow[d, hook] & X^n \arrow[d] \\ D_\alpha^n \arrow[r, "\Phi_\alpha"'] & X^{n+1} \end{tikzcd}$$

Indeed Φ_α is determined by φ_α as a property of the pushout. Collectively, we identify the attaching of all n-cells as the following pushout:

$$\begin{tikzcd} \coprod_{\alpha \in I_n} S_\alpha^{n-1} \arrow[r, "\coprod_{\alpha \in I_n} \varphi_\alpha"] \arrow[d, hook] & X^n \arrow[d] \\ \coprod_{\alpha \in I_n} D_\alpha^n \arrow[r, "\coprod_{\alpha \in I_n} \Phi_\alpha"'] & X^{n+1} \end{tikzcd}$$

The final CW complex X is then obtained by the union with the weak topology. But this is just the explicit description of the colimit of the n-skeletons.

Proposition 1.2.2 Let X, Y be spaces and $A \subseteq X$ a subspace of X. Let $f : A \to Y$ be a map. Then the adjunction space $X \amalg_f Y$ is a pushout of f and $i : A \to X$ in $\mathbf{Top}$.

Proposition 1.2.3 Let X, Y be spaces with chosen base point x_0 and y_0 respectively. Then the wedge product

$$X \vee Y = X \amalg_f Y$$

is an adjunction space with $Z = \{x_0\}$ and map $f : Z \to Y$ defined by $f(x_0) = y_0$.

Definition 1.2.4 (Mapping Cone) Let X, Y be spaces. Let $f : X \to Y$ be a map. Define the mapping cone of f to be

$$C_f = CX \amalg_f Y = \frac{CX \amalg Y}{\sim}$$

where the relation is generated by $(x, 0) \sim f(x)$ for $x \in X$, and CX is the unreduced cone of X.

1.3 Categories of CW Complexes

Definition 1.3.1 (The Category of CW Complexes)
Define the category of CW complexes **CW** to consist of the following data.

- The objects are CW complexes.
- For X and Y two CW complexes, the morphisms

$$\text{Hom}_{\mathbf{CW}}(X, Y) = \{f : X \to Y \mid f \text{ is a cellular map }\}$$

  are the continuous maps from X to Y.
- Composition is defined as the composition of functions.

Definition 1.3.2 (The Category of Pointed CW Complexes)
Define the category of CW complexes $\mathbf{CW}_*$ to consist of the following data.

- The objects are pairs where (X, x_0) where X is a CW complex and $x_0 \in X^0$ lies in the zero skeleton.
- For (X, x_0) and (Y, y_0) two pointed CW complexes, the morphisms are given by

$$\text{Hom}_{\mathbf{CW}_*}(X, Y) = \{f : X \to Y \mid f \text{ is a cellular map and } f(x_0) = y_0\}$$

- Composition is defined as the composition of functions.

Definition 1.3.3 (The Category of Pairs of CW Complexes)
Define the category of pairs of CW complexes $\mathbf{CW}^2$ to consist of the following data.

- The objects are pairs (X, A) where X is a CW complex and A is a subcomplex of X.
- For (X, A) and (Y, B) two pairs of CW complexes, the morphisms are given by

$$\text{Hom}_{\mathbf{CW}^2}((X, A), (Y, B)) = \{f : X \to Y \mid f(A) \subseteq B, f \text{ and } f|_A \text{ are a celluar maps }\}$$

- Composition is defined as the composition of functions.

2 Singular Homology

2.1 Singular n-Simplexes and Singular Homology

There are a few problems with simplicial homology. In particular, not every space has a $Delta$-complex structure. We would want to extend this definition to any space, with or without Δ-complex structures. Moreover, we have not yet shown that simplicial homology is independent of the choice of Δ-complex structures. Maps between Δ-complex structures may not necessarily define a map between its homology groups.

We therefore want a better version of homology that does all of this, and in particular, to allow any space to have a well defined homology groups.

Definition 2.1.1 (Singular n-Simplexes) A singular n-simplex in a topological space X is a continuous map $\sigma : \Delta^n \to X$ where Δ^n is an n-simplex.

We say that these are singular because we allow potential deformations of the faces.

It is clear that every Δ-complex consists of n-simplex so the inclusion map of the n-simplex to the Δ-complex defines a singular n-simplex. In fact by definition, none of these n-simplex are "singular" in the sense that some of their faces are degenerate.

Definition 2.1.2 (The Group of Singular n-Chains) Let X be a topological space. Define the group of singular n-chains on X to be the free abelian group

$$C_n(X) = \mathbb{Z}\langle \sigma : \Delta^n \to X \mid \sigma \text{ is a singular } n \text{ simplex} \rangle$$

on the set of singular n-chains.

This definition is reminiscent of that of Δ-sets and Δ-complexes. The important point is that to every set we can associate a free group on the set. As a set itself there is no algebraic invariants to study. But once we enrich it to include formal linear combinations, we obtain a group structure.

Definition 2.1.3 (The Boundary Operator) Let X be a topological space. Define the boundary operator $\partial_n : C_n(X) \to C_{n-1}(X)$ to be the homomorphism given by

$$\partial_n(\sigma) = \sum_{k=0}^{n} (-1)^k \sigma|_{\partial_i \Delta^n}$$

The same proof as in proposition 2.2.3 gives the following.

Proposition 2.1.4 Let X be a space. The family of abelian groups $C_n(X)$ of a simplicial complex and the boundary operator ∂ forms a chain complex.

Proof Let $\sigma \in C_{n+1}(X)$. Then we have that

$$(\partial_n \circ \partial_{n+1})(\sigma) = \partial_n \left(\sum_{j=0}^{n+1} (-1)^j \sigma|_{\partial_j \Delta^{n+1}} \right) = \sum_{i=0}^{n} \sum_{j=0}^{n+1} (-1)^{i+j} \left(\sigma|_{\partial_j \Delta^{n+1}} \right) |_{\partial_i \Delta^n}$$

Fix a pair $0 \leq i < j \leq n+1$. By definition 2.1.5, $A = (-1)^{i+j} \left(\sigma|_{\partial_j \Delta^{n+1}} \right) |_{\partial_i \Delta^n}$ and $B = (-1)^{i+j} \left(\sigma|_{\partial_i \Delta^{n+1}} \right) |_{\partial_{j-1} \Delta^n}$ cancel out. Moreover every summand is of the form A or B and not both so that the sum vanishes. In other words, we have that

$$\begin{aligned}
&\sum_{i=0}^{n}\sum_{j=0}^{n+1}(-1)^{i+j}\left(\sigma|_{\partial_j\Delta^{n+1}}\right)|_{\partial_i\Delta^n} \\
&= \sum_{0\leq i<j\leq n+1}(-1)^{i+j}\left(\sigma|_{\partial_j\Delta^{n+1}}\right)|_{\partial_i\Delta^n} + \sum_{0\leq j<i\leq n}(-1)^{i+j}\left(\sigma|_{\partial_j\Delta^{n+1}}\right)|_{\partial_i\Delta^n} \\
&= \sum_{0\leq i<j\leq n+1}(-1)^{i+j}\left(\sigma|_{\partial_i\Delta^{n+1}}\right)|_{\partial_{j-1}\Delta^n} + \sum_{0\leq j<i\leq n}(-1)^{i+j}\left(\sigma|_{\partial_j\Delta^{n+1}}\right)|_{\partial_i\Delta^n} \\
&= 0
\end{aligned}$$

We conclude that $(C_\bullet, \partial_\bullet)$ forms a chain complex. ■

Definition 2.1.5 (The Singular Chain Complex) Let X be a space. Define the singular chain complex $(C_\bullet(X), \partial_\bullet)$ to be the chain complex given by the following data.

- The nth term is given by
$$C_n(X) = \mathbb{Z}\langle \sigma : \Delta^n \to X \mid \sigma \text{ is a singular } n \text{ simplex}\rangle$$
the group of singular n-chains of X.
- The boundary operator $\partial_n : C_n(X) \to C_{n-1}(X)$ is given by the formula
$$\partial_n(\sigma) = \sum_{k=0}^{n}(-1)^k\sigma|_{\partial_i\Delta^n}$$

Definition 2.1.6 (The Singular Homology Group) Let X be a topological space. Define the nth singular homology group of X to be

$$H_n(X) = H_n(C_\bullet(X), \partial_\bullet) = \frac{\ker(\partial_n : C_n(X) \to C_{n-1}(X))}{\operatorname{im}(\partial_{n+1} : C_{n+1}(X) \to C_n(X))}$$

One can imagine that in order to deduce the homology groups of a space, the computations involved are highly non-trivial. For instance, one has first work out how the boundary maps are defined in order to deduce their kernel and images. But the set of singular n-simplexes of a space is a much larger set than that of Δ-complexes. When the dimension of the space X (whatever this means) is larger, one can also imagine that the singular homology groups $H_n(X)$ are non-trivial in most of $n \in \mathbb{N}$. We will see how to compute them in the next section.

Example 2.1.7 Let $X = *$ be a point. Then the homology of the one point space is

$$H_n(*) \cong \begin{cases} \mathbb{Z} & \text{if } n = 0 \\ 0 & \text{otherwise} \end{cases}$$

Proof For each $n \geq 0$, there is a unique singular n-simplex, $c_n : \Delta^n \to *$ the constant map. Thus the singular chain complex becomes

$$\cdots \longrightarrow \mathbb{Z} \xrightarrow{\partial_2} \mathbb{Z} \xrightarrow{\partial_1} \mathbb{Z} \longrightarrow 0$$

Now notice that

$$\begin{aligned}
\partial_n(c_n) &= \sum_{i=0}^{n}(-1)^n c_n|_{\partial_i\Delta^n} \\
&= \sum_{i=0}^{n}(-1)^n c_{n-1} \\
&= \begin{cases} c_{n-1} & \text{if } n > 0 \text{ is even} \\ 0 & \text{otherwise} \end{cases}
\end{aligned}$$

This means that ∂_n is an isomorphism when n is even, ∂_n is the zero map when n is odd. When $n \neq 0$ is even, we have that

$$H_n(*) = \frac{\ker(\partial_n)}{\operatorname{im}(\partial_{n+1})} = \frac{\{0\}}{\{0\}} = 0$$

When n is odd, we have that

$$H_n(*) = \frac{\ker(\partial_n)}{\operatorname{im}(\partial_{n+1})} = \frac{\mathbb{Z}}{\mathbb{Z}} = 0$$

Finally when $n = 0$, we have that $H_0(*) = \mathbb{Z}$. ■

For now, we are concerned with some immediate consequences of this definition, such as functoriality.

Proposition 2.1.8 Let $f : X \to Y$ be a continuous map. Then f induces a chain map

$$f_* : H_n(X) \to H_n(Y)$$

defined by $[\sigma] \mapsto [f \circ \sigma]$ for each $\sigma : \Delta^n \to X$ a singular n-simplex. This map satisfies

- $\mathrm{id}_* = \mathrm{id}_{H_n(X)}$
- If $g : Y \to Z$ is another continuous map, then $(g \circ f)_* = g_* \circ f_*$

Proof Let $\sigma : \Delta^n \to X$ be a singular n-simplex in X. Then $f \circ \sigma : \Delta^n \to Y$ is a singular n-simplex in Y by continuity of f. We can linearly extend this to a group homomorphism $f_n : C_n(X) \to C_n(Y)$. The collection of these group homomorphisms lead to a chain map $f_* : C_\bullet(X) \to C_\bullet(Y)$: Indeed we have that

$$\begin{aligned}
f_{n-1} \circ \partial_n(\sigma) &= f_{n-1}\left(\sum_{k=0}^{n}(-1)^k \sigma|_{\partial_i \Delta^n}\right) \\
&= \sum_{k=0}^{n}(-1)^k f_{n-1}\left(\sigma|_{\partial_i \Delta^n}\right) \\
&= \sum_{k=0}^{n}(-1)^k \left(\sigma|_{\partial_i (f_n(\Delta^n))}\right) \\
&= \partial_n(f_n(\sigma))
\end{aligned}$$

which shows that f is a chain map. By lemma 1.1.4, this yields a group homomorphism $f_n : H_n(X) \to H_n(Y)$ on each degree.

It is clear that the chain map $\mathrm{id}_* : C_\bullet(X) \to C_\bullet(Y)$ is the identity and so it also descends to the identity map on homology groups. The second property also follows immediately from the construction above. ■

Proposition 2.1.9 Let X be a topological space and that $\{X_\alpha | \alpha \in I\}$ is the path components of X. Then

$$H_n(X) = \bigoplus_{\alpha \in I} H_n(X_\alpha)$$

Proof Let $\sigma : \Delta^n \to X$ be a singular n-simplex. Then its image is path-connected and therefore lies entirely in one of the X_α. This means that

$$C_n(X) = \bigoplus_{\alpha \in I} C_n(X_\alpha)$$

Moreover, the boundary of σ is a linear combination of $(n-1)$ simplices which all lie in X_α. This means that the chain complex splits into

$$C_\bullet(X) = \bigoplus_{\alpha \in I} C_\bullet(X_\alpha)$$

This decomposition therefore passes down to cycles, boundaries and homology. ■

2.2 Relation to the Low Degree Homotopy Groups

We follow up with a geometric interpretation of H_0 and H_1. This will make calculations slightly easier, especially since we are able to relate H_1 with the fundamental group, which we have seen various examples of in Algebraic Topology 1

Definition 2.2.1 (Homologous Elements) Let X be a topological space. Let $x, y \in C_n(X)$. We say that x and y are homologous if there exists $z \in C_{n+1}(X)$ such that $\partial_{n+1}(z) = x - y$.

Intuitively, two elements are homologous if they bound some sort of higher dimensional singular complex. In terms of Δ-complexes and CW complexes, two n-chains are homologous if they form the boundary of an $(n+1)$-chain.

Lemma 2.2.2 Let X be a path connected space. Then the 0th homology is the integers

$$H_0(X) \cong \mathbb{Z}$$

Proof Define a map $\deg : C_0(X) \to \mathbb{Z}$ by $\deg(x) = 1$ for every x in the generator of $C_0(X)$ and extend it linearity so that deg is a group homomorphism. Firstly $\deg$ is surjective since X is non-empty. Indeed there exists $x \in X$ with its image a generator in $\mathbb{Z}$.

Now we show $B_0(X) \subseteq \ker(\deg)$. Let $\gamma : \Delta^1 \to X$. Then

$$\begin{aligned} \deg(\partial_1\gamma) &= \deg(\gamma(1) - \gamma(0)) \\ &= \deg(\gamma(1)) - \deg(\gamma(0)) \\ &= 1 - 1 \\ &= 0 \end{aligned}$$

Thus we are done. Now we show that $\ker(\deg) \subseteq B_0(X)$. Suppose that $L = \sum_{x \in X} \lambda_x \cdot x \in \ker(\deg)$ for $\lambda_x \in \mathbb{Z}$ and finitely many non-zero. Then we have the following:

$$L = \sum_{\substack{x \in X \\ \lambda_x \geq 0}} \lambda_x \cdot x - \sum_{\substack{y \in X \\ \lambda_y < 0}} (-\lambda_y) \cdot y$$

$$\begin{aligned} 0 = \deg(L) &= \deg\left(\sum_{\substack{x \in X \\ \lambda_x \geq 0}} \lambda_x \cdot x - \sum_{\substack{y \in X \\ \lambda_y < 0}} (-\lambda_y) \cdot y \right) \\ &= \sum_{\substack{x \in X \\ \lambda_x \geq 0}} \lambda_x - \sum_{\substack{y \in X \\ \lambda_y < 0}} (-\lambda_y) \end{aligned}$$

This means that we can pair up the positives and the negatives so that

$$L = \sum (x_i - y_i)$$

for $x_i \in X$ with positive coefficient and $y_i \in X$ for negative coefficients.

Now observe the following: If $\gamma : [0,1] = \Delta^1 \to X$ is a singular 1-simplex with $\gamma(0) = x$ and $\gamma(1) = y$ with $x, y \in C_0(X)$, then

$$\partial_1(\gamma) = \gamma|_{\partial_0\Delta^1} - \gamma|_{\partial_1\Delta^1} = \gamma(1) - \gamma(0) = y - x$$

Since X is path connected, for any x_i, y_i, there exists $\gamma_i : [0,1] = \Delta^1 \to X$ such that $\gamma_i(0) = x_i$,

$\gamma_i(1) = y_i$. Then

$$\begin{aligned} L &= \sum (x_i - y_i) \\ &= \sum \partial_1 \gamma_i \\ &= \partial_1 \left(\sum_i \gamma_i \right) \end{aligned}$$

Thus $L \in B_0(X)$. Combining the fact that $\deg$ is surjective and $B_0(X) = \ker(\deg)$, we obtain

$$H_0(X) \cong \mathbb{Z}$$

by the first isomorphism theorem and thus we are done. ■

Using the lemma, we can now interpret $H_0(X)$ as the free group on the path components of X.

Corollary 2.2.3 Let X be a space. Then

$$H_0(X) \cong \mathbb{Z}\pi_0(X)$$

Proof For any space X, $\pi_0(X)$ is the path components of X. We know from proposition 3.2.2 that the zero homology of each path connected components is $\mathbb{Z}$. Proposition 3.1.7 shows that we can split the homology of X into direct sum of homology of path connected components. This leaves us with the claim. ■

The remainder of this section is dedicated to the first homology group $H_1(X)$ and its relation to the fundamental group $\pi(X, x)$. In fact we almost have a well defined map

$$h_1 : \pi_1(X, x) \to H_1(X)$$

given as follows. Start with a loop $\gamma : \Delta^1 \to X$ in $\pi_1(X, x)$. It follows that $\partial_1(\gamma) = \gamma(1) - \gamma(0) = x - x = 0$ so that γ is a cycle in $C_1(X)$. Since $\pi_1(X, x)$ is defined as an equivalence class of homotopic loops, we would like h_1 to send an equivalence class to $H_1(X)$ instead of just a loop itself. For this, we need to show that any two homotopic loops map to the same element is homologous.

Lemma 2.2.4 Let γ_1, γ_2 be two paths in a space X that are homotopic relative to their end points. Then γ_1 and γ_2 are homologous elements of $C_1(X)$.

Proof Suppose that $H : I \times I \to X$ is the homotopy between γ_1 and γ_2 relative to end points x and y. Let c_x be the constant loop at x and vice versa for c_y. Define $\gamma(t) = H(t, t)$. Then $\gamma_1 \cdot c_y \cdot \overline{\gamma} = 0$ which means that it is the boundary of some 2-chain, say σ_1. Similarly, $c_x \cdot \gamma_2 \cdot \overline{\gamma} = 0$ and so it is the boundary of a 2-chain, say σ_2. We then have that

$$\begin{aligned} \partial_2(\sigma_2 - \sigma_1) &= \gamma_2 - \gamma + c_x - c_y + \gamma - \gamma_1 \\ &= (\gamma_2 - \gamma_1) + (c_x - c_y) \end{aligned}$$

Our goal is to show that $c_x - c_y \in B_1(X)$ so that γ_1 and γ_2 are homologous via the 2-chain $\sigma_2 - \sigma_1$.

Let $\sigma : \Delta^2 \to X$ be the constant map with value x. Then

$$\partial_2(\sigma) = c_x - c_x + c_x = c_x$$

shows that $c_x \in B_1(X)$. This is the same for c_y and so every constant path on X lie in $B_1(X)$. ■

Thus now we have a map of sets

$$h_1 : \pi_1(X, x) \to H_1(X)$$

we still need to show that it is a group homomorphism. The following lemma will help in the proof.

Lemma 2.2.5 Let γ_1, γ_2 be two paths in X with $\gamma_1(1) = \gamma_2(0)$. Then $\gamma_1 \cdot \gamma_2$ is homologous to $\gamma_1 + \gamma_2$. Moreover, $\overline{\gamma}$ is homologous to $-\gamma$.

Proof It is clear that $\gamma_1, \gamma_2, \gamma_1 \cdot \gamma_2$ form the boundary of a 2-simplex, say $\sigma = [v_0, v_1, v_2]$ since $\gamma_1 \cdot \gamma_2 \cdot \overline{\gamma_1 \cdot \gamma_2} = 0$. Now project v_1 orthogonally down to the face $[v_0, v_2]$ to get a new two simplex

$$\sigma : [v_0, v_1, v_2] \to [v_0, v_2] \to X$$

Then we have $\partial_2(\sigma) = \gamma_1 + \gamma_2 - \gamma_1 \cdot \gamma_2$ which shows that $\gamma_1 + \gamma_2$ and $\gamma_1 \cdot \gamma_2$ are homologous.

Now we have that $\gamma + \overline{\gamma}$ is homologous to $\gamma \cdot \overline{\gamma}$, and this is homologous to the trivial loop. This means that $\overline{\gamma}$ is homologous to $-\gamma$. ■

This gives the following proposition.

Proposition 2.2.6 Let X be a topological space. The map

$$h_1 : \pi_1(X, x) \to H_1(X)$$

defined by $h_1([\gamma]) = [\gamma]$ for $[\gamma] \in \pi_1(X, x)$ is a group homomorphism.

Proof We have shown from lemma 3.2.4 that every constant path on X lie in $B_1(X)$. Then $h_1([c_x]) = (0 + B_1(X)) \in H_1(X)$ implies that h_1 maps units to units. Now let γ_1, γ_2 be two loops based at x and $\gamma_1 \cdot \gamma_2$ be their concatenation. Our goal is to show that $h_1([\gamma_1] \cdot [\gamma_2]) = h_1([\gamma_1]) + h_1([\gamma_2])$. This amounts to showing that $\gamma_1 \cdot \gamma_2$ is homologous to $\gamma_1 + \gamma_2$. Then by the above lemma, we are done. ■

In general, $h_1 : \pi_1(X, x) \to H_1(X)$ is a map to an abelian group $H_1(X)$. However as we have seen in Algebraic Topology 1, not every space has an abelian fundamental group. However, if we forcefully abelianize the fundamental group, we in fact obtain an isomorphism.

Theorem 2.2.7 Let X be a non-empty path connected topological space. Then there is an isomorphism

$$\pi_1(X, x)^{\text{ab}} \cong H_1(X)$$

Proof Since X is path connected, for every $y \in X$, we can once and for all, choose a path η_y from x to y. Given any path $\gamma : \Delta^1 \to X$, we associate a loop based at x, as the following concatenation:

$$g(\gamma) = \eta_{\gamma(0)} \cdot \gamma \cdot \eta_{\gamma(1)}^{-1}$$

We can extend this map linearly to obtain a homomorphism

$$g : Z_1(X) \subseteq C_1(X) \to \pi_1(X, x)^{\text{ab}}$$

Now we want $g(b) = 0$ for any boundary $b \in B_1(X)$. Now notice that for a singular 2-simplex with boundary $\gamma_1 : I \to [v_0, v_1], \gamma_2 : I \to [v_1, v_2], \gamma_3 : I \to [v_0, v_2]$, we have that $\gamma_1 \cdot \gamma_2$ is homotopic relative to their end points. It follows that for $\partial_2(\sigma) \in B_1(X)$, we have

$$\begin{aligned} g(\partial_2(\sigma)) &= g(\gamma_1 + \gamma_2 - \gamma_3) \\ &= g(\gamma_1) + g(\gamma_2) - g(\gamma_3) \\ &= [\eta_{v_0} \cdot \gamma_1 \cdot \eta_{v_1}^{-1}] + [\eta_{v_1} \cdot \gamma_2 \cdot \eta_{v_2}^{-1}] + [\eta_{v_2} \cdot \overline{\gamma_3} \cdot \eta_{v_0}^{-1}] \\ &= [\eta_{v_0} \cdot \gamma_1 \cdot \gamma_2 \cdot \overline{\gamma_3} \cdot \eta_{v_0}^{-1}] \\ &= [\eta_{v_0} \eta_{v_0}^{-1}] \\ &= 0 \end{aligned}$$

This means that $\overline{g} : H_1(X) \to \pi_1(X, x)^{\text{ab}}$ is well defined.

It remains to show that the two composites are the identity. Let $[gamma] \in \pi_1(X,x)^{\text{ab}}$. Then we have

$$\begin{aligned}\overline{g}(\overline{h}_1([\gamma])) &= \overline{[\gamma]} \\ &= [\eta_x \cdot \gamma \cdot \eta_x^{-1}] \\ &= [\eta_x] + [\gamma] + [\eta_x^{-1}] \\ &= [\gamma]\end{aligned}$$

Thus $\overline{g} \circ \overline{h}_1 = \text{id}$. Now let $L = \sum \lambda_\gamma \gamma \in Z_1(X)$ by a 1-cycle where $\lambda_\gamma \in \mathbb{Z}$. By replacing $-\gamma$ by $\overline{\gamma}$ if necessary (Lemma 3.2.5), we may assume that $\lambda_\gamma \in \mathbb{N} \setminus \{0\}$. Relabelling gives

$$L = \sum_{i=1}^n \gamma_i$$

where γ_i can possibly repeat. If γ_1 is not a loop, then there must exist $i > 1$ such that $\gamma_1(1) = \gamma_i(0)$. Replacing $\gamma_1 + \gamma_i$ by the concatenation $\gamma_1 \cdot \gamma_i$ (By lemma 3.2.5) and doing induction reduces the claim for $L = \gamma$ a single loop, say based at y. In this case we have that

$$\begin{aligned}\overline{h_1}(\overline{g}([\gamma])) &= [\eta_y \cdot \gamma \cdot \eta_y^{-1}] \\ &= [\eta_y] + [\gamma] - [\eta_y] \\ &= [\gamma]\end{aligned}$$

and this completes the proof. ■

This gives a rather nice interpretation of the first homology group: It is the abelianization of the fundamental group. Intuitively, since $H_1(X)$ is abelian, it makes detecting differences in this invariant easier, when compared to the non-abelian $\pi_1(X,x)$, which also depends on base point.

We have an immediate application based on calculations of the fundamental group.

Corollary 2.2.8 If X is simply connected then $H_1(X) = 0$.

Proof In this case the fundamental group is trivial and by the above theorem, we have that $H_1(X) = 0$. ■

2.3 Reduced Homology

Using corollary 3.2.3, we see that path-connected spaces will have non-trivial 0th homology group. This motivates the minor modification into reduced homology since the intuition should show that homology groups of single points should be equal to 0.
We shall see in later chapters that this also simplifies the statements of homology in many cases.

Definition 2.3.1 (The Augmented Chain Complex) Let X be a space. Define the augmented chain complex $(\widetilde{C}_\bullet(X), \partial_\bullet)$ of X to be the chain complex given by

$$\cdots \xrightarrow{\partial_{n+1}} C_n(X) \xrightarrow{\partial_n} \cdots \xrightarrow{\partial_2} C_1(X) \xrightarrow{\partial_1} C_0(X) \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0$$

where $\varepsilon : C_0 \to \mathbb{Z}$ is the map given by

$$\varepsilon\left(\sum_{i\in I} n_i\sigma_i\right) = \sum_{i\in I} n_i$$

Definition 2.3.2 (Reduced Homology) Let X be a space. Define the nth reduced homology group of X to be

$$\widetilde{H}_n(X) = H_n(\widetilde{C}_\bullet(X), \partial_\bullet)$$

It is clear that for $n \geq 1$, singular homology and reduced homology gives the same homology groups, but not for $n = 0$.

Lemma 2.3.3 Let X be a topological space. Then for $n \geq 1$, the homology and the reduced homology are equal. This means that

$$\widetilde{H}_n(X) \cong H_n(X)$$

for all $n \geq 1$.

Proof The chain complex is entirely the same for $n \geq 1$ so their homology groups will also be the same. ∎

When $n = 0$, we must have that

$$\tilde{H}_0(X) \cong \frac{\ker(\varepsilon)}{\operatorname{im}(\partial_1)}$$

There is in fact an alternate definition of the reduced homology groups, given as follows.

Proposition 2.3.4 Let X be a space. Let $\pi : X \to *$ be the unique map to the one point space. Then there is a split exact sequence

$$0 \longrightarrow \widetilde{H}_0(X) \xrightarrow{f} H_0(X) \xrightarrow{\pi_*} \mathbb{Z} \longrightarrow 0$$

where f is the inclusion map.

Proof We first show that f is well defined. Notice that $\ker(\varepsilon) \leq C_0(X) \twoheadrightarrow \frac{C_0(X)}{\operatorname{im}(\partial_1)} = H_0(X)$ so that there is a canonical map $\ker(\varepsilon) \to H_0(X)$ given by $x \mapsto [x]$. Then kernel of this map is precisely $\ker(\varepsilon) \cap \operatorname{im}(\partial_1) = \operatorname{im}(\partial_1)$. Since $\operatorname{im}(\partial_1) \leq \ker(\varepsilon)$ our map descends to a well defined map $\widetilde{H}_0(X) = \frac{\ker(\varepsilon)}{\operatorname{im}(\partial_1)}$ which is precisely the inclusion map. The inclusion map is injective and π is surjective since it is a retract. We are left to show that $\operatorname{im}(f) = \ker(\pi_*)$.

Now π_* sends all generators of $H_0(X)$ to the unique element of the one point set, hence π_* is the map given by $[\sum_{k=0}^{n} m_k x_k] \mapsto \sum_{k=1}^{n} m_k$. Since ε has the exact same effect as π_* we conclude that $\ker(\varepsilon) = \ker(\pi_*)$.

Finally since the third term of the short exact sequence is free, we conclude that the short exact sequence splits. ∎

Note that the upcoming two theorems: homotopy invariance and Mayer Vietoris also holds for reduced homology.

2.4 Homotopy Invariance

The goal of this subsection is to establish homotopy invariance for singular homology. This is not at all obvious compared to that of fundamental groups. We will split the proof into the part involving topology, and the part involving algebra. In fact, we have already completed the proof on the algebra side in chapter 1. This is the notion of chain homotopy. In category terms they represent homotopy between morphisms of objects. Therefore the notion of chain maps is important in proving any theorem involving homotopies.

For the topological side, we define the prism operator on Δ-sets first.

Definition 2.4.1 (Prism Operator) Let $\Delta^n = [v_0, \dots, v_n]$ be an n-simplex. Define the prism operator by

$$P(\Delta^n) = \sum_{i=0}^{n} (-1)^i [v_{00}, \dots, v_{i0}, v_{i1}, \dots, v_{n1}] \in C_{n+1}(\Delta^n \times [0,1])$$

where v_{ij} denotes the ith vertex of the jth Δ^n simplex for $0 \leq i \leq n$ and $0 \leq j \leq 1$.

As we have already seen that chain homotopies produce equal maps, our goal is now to establish a chain homotopy from two homotopic maps $f, g : X \to Y$ of spaces. Given a homotopy $H : X \times I \to Y$ of f and g and a simplex $\sigma : \Delta^n \to X$ in X, we can compose them to obtain a map:

$$H \circ (\sigma \times \text{id}) : \Delta^n \times I \to Y$$

so that we now have a homotopy between two singular n-simplexes, namely $f \circ \sigma : \Delta^n \to Y$ and $g \circ \sigma : \Delta^n \to Y$. Recalling from the definition of chain homotopies, we need to produce an $(n+1)$-simplex from this datum. This is done by considering the prism in between $f \circ \sigma$ and $g \circ \sigma$. In particular, $\sigma \times I$ is a prism, and we could divide up the prism into simplexes. This is precisely the content of the Prism operator.

The sign changes in the prism operator is defined so that the boundary of the prism produces the following equality, which is reminiscent with that of chain homotopies.

Lemma 2.4.2 For every $n \geq 0$, we have in $C_n(\Delta^n \times [0,1])$ that

$$\partial P(\Delta^n) = [v_{01}, \dots, v_{n1}] - [v_{00}, \dots, v_{n0}] - P(\partial \Delta^n)$$

Proof We have that

$$\begin{aligned}\partial_{n+1} P(\Delta^n) = &\sum_{j \leq i} (-1)^{i+j} [v_{00}, \dots, \hat{v}_{j0}, \dots, v_{i0}, v_{i1}, \dots, v_{n1}] \\ &+ \sum_{j \geq i} (-1)^{i+j+1} [v_{00}, \dots, v_{i0}, v_{i1}, \dots, \hat{v}_{j1}, \dots, v_{n1}]\end{aligned}$$

Notice that for $i = j$, we get

$$\sum_{i=0}^{n} [v_{00}, \dots, v_{(i-1)0}, v_{i1}, \dots, v_{n1}] - \sum_{i=0}^{n} [v_{00}, \dots, v_{i0}, v(i+1)1, \dots, v_{n1}]$$

All but two of which cancel out, leaving us with

$$[v_{01}, \dots, v_{n1}] - [v_{00}, \dots, v_{n0}]$$

For $i \neq j$, apply the prism operator P to each face $[v_0, \dots, \hat{v}_j, \dots, v_n]$ of Δ^n to get

$$\begin{aligned}P([v_0, \dots, \hat{v}_j, \dots, v_n]) = &\sum_{i<j} (-1)^i [v_{00}, \dots, v_{i0}, v_{i1}, \dots, \hat{v}_{j1}, \dots, v_{n1}] \\ &+ \sum_{j<i} (-1)^{i+1} [v_{00}, \dots, \hat{v}_{j0}, \dots, v_{i0}, v_{i1}, \dots, v_{n1}]\end{aligned}$$

Taking the alternating sum over all j, we get

$$\begin{aligned}P(\partial_n \Delta^n) = &\sum_{i<j} (-1)^{i+j} [v_{00}, \dots, v_{i0}, v_{i1}, \dots, \hat{v}_{j1}, \dots, v_{n1}] \\ &+ \sum_{j<i} (-1)^{i+j+1} [v_{00}, \dots, \hat{v}_{j0}, \dots, v_{i0}, v_{i1}, \dots, v_{n1}]\end{aligned}$$

This is precisely the negative terms of of $\partial_{n+1} P(\Delta^n)$ yet to be accounted for (terms for which $i \neq j$) Combining the results concludes the proof. ∎

We can now prove homotopy invariance of the singular homology groups.

Theorem 2.4.3 (Homotopy Invariance) Suppose $f, g : X \to Y$ are homotopic continuous maps. Then they induce the same homomorphism

$$f_n = g_n : H_n(X) \to H_n(Y)$$

In particular, if X and Y are homotopy equivalent then $H_n(X) \cong H_n(Y)$.

Proof Suppose that $H : X \times I \to Y$ is a homotopy from f to g. Let $\sigma : \Delta^n \to X$ be a singular n-simplex in X. Consider $H \circ (\sigma \times \text{id}) : \Delta^n \times [0,1] \to Y$. This map induces a map on $(n+1)$-chains:

$$(H \circ (\sigma \times \text{id}))_* : C_{n+1}(\Delta^n \times [0,1]) \to C_{n+1}(Y)$$

Then define a chain homotopy $\eta_n : C_n(X) \to C_{n+1}(Y)$ by

$$\eta_n(\sigma) = (H \circ (\sigma \times \text{id}))_*(P(\Delta^n))$$

Indeed the calculation

$$\begin{aligned}
\partial(\eta_n(\sigma)) &= \partial(H \circ (\sigma \times \text{id})_*(P(\Delta^n))) && \text{(Definition of } \eta_n\text{)} \\
&= H \circ (\sigma \times \text{id})_*(P(\Delta^n)) && \text{(Chain map)} \\
&= H \circ (\sigma \times \text{id})_*([v_{01}, \dots, v_{n1}] - [v_{00}, \dots, v_{n0}] - P(\partial\Delta^n)) && \text{(By the above lemma)} \\
&= g_*(\sigma) - f_*(\sigma) - \eta_{n-1}(\partial(\sigma))
\end{aligned}$$

shows that η_n satisfies the chain homotopy equation. Since chain homotopy induces the same map in homology, we are done. ■

The following result is immediate from homotopy invariance. In fact, it is the important result that we will use all the time. For example, by deformation retracting a space into a known subspace, we can deduce at once its singular homology groups.

Corollary 2.4.4 Let X and Y be homotopy equivalent, then $H_n(X) \cong H_n(Y)$ are isomorphic.

Proof Suppose that the homotopy equivalence is given by $f : X \to Y$ and $g : Y \to X$. That is, we have $f \circ g \simeq \text{id}_Y$ and $g \circ f \simeq \text{id}_X$. Then by propsition 3.1.6 and homotopy invariance, we have that

$$f_* \circ g_* = (f \circ g)_* = (\text{id}_Y) = \text{id}$$

and similarly $g_* \circ f_* = \text{id}$. Thus g_* and f_* are inverses of each other and so $H_n(X) \cong H_n(Y)$. ■

Homotopy invariance is also true for reduced homology.

Example 2.4.5 Let X be a contractible space. Then

$$H_n(X) \cong \begin{cases} \mathbb{Z} & \text{if } n = 0 \\ 0 & \text{otherwise} \end{cases}$$

Proof Follows from the homology of one point space and homotopy invariance. ■

2.5 Mayer-Vietoris Sequence

Seifert-van Kampen theorem allows the fundamental group of a space to be computed by considering appropriate subspaces. There is a similar method for homology provided by the Mayer-Vietoris sequence.

The Mater-Vietoris sequence is another powerful for breakdown spaces into subspaces in order to compute homology groups. We will make use of the notion of Barycentric subdivisions to proof the theorem.

We can transfer the definition of barycentric subdivision simply by pushing forward.

Definition 2.5.1 (Barycentric Subdivision of Singular Simplices) Let X be a space and $\sigma : \Delta^n \to X$ a singular n-simplex. Let $\sigma_* : C_n(\sigma) \to C_n(X)$ be the induced map given on the basis by $[v_0, \dots, v_n] \mapsto \sigma|_{[v_0,\dots,v_n]}$. Define the barycentric subdivision of σ to be the n-chain

$$S(\sigma) = \sigma_*(S^\Delta(\Delta^n))$$

Extend it to a map $S : C_n(X) \to C_n(X)$ defined on the basis.

Proposition 2.5.2 Let $n \in \mathbb{N}$. Define $T(\Delta^n) \in C_{n+1}(\Delta^n \times I)$ as follows.

- $T^\Delta(\Delta^0) = \mathcal{B}(\Delta^0) = [B(\Delta^0), v] \in C_1(\Delta^0 \times I)$
- For $n > 0$, define $T^\Delta(\Delta^n) = \mathcal{B}(\Delta^n \times \{0\}) - \mathcal{B}(T(\partial(\Delta^n \times \{0\}))) \in C_{n+1}(\Delta^n \times I)$

Then we have

$$\partial(T^\Delta(\Delta^n)) + T(\partial(\Delta^n \times \{0\})) = \Delta^n \times \{0\} - S(\Delta^n \times \{1\})$$

Proof It is trivially true for $n = 0$. We then induct on n. Suppose that the previous cases are true. We have that

$$\begin{aligned}
\partial(T(\Delta^n)) &= \partial(\mathcal{B}\Delta_0^n) - \partial(\mathcal{B}(T(\partial\Delta_0^n))) \\
&= \Delta_0^n - \mathcal{B}(\partial\Delta_0^n) - T(\partial\Delta_0^n) + \mathcal{B}(\partial T(\partial\Delta_0^n)) && \text{(Lemma 4.2.4)} \\
&= \Delta_0^n - T(\partial\Delta_0^n) + \mathcal{B}(\partial T(\partial\Delta_0^n) - \partial\Delta_0^n) \\
&= \Delta_0^n - T(\partial\Delta_0^n) - \mathcal{B}(S(\partial\Delta_1^n) + T(\partial^2\Delta_0^n)) && \text{(By induction)} \\
&= \Delta_0^n - T(\partial\Delta_0^n) - S(\Delta_1^n)
\end{aligned}$$

■

Proposition 2.5.3 Let X be a space. Then the barycentric subdivision map $S : C_n(X) \to C_n(X)$ is a chain map that is chain homotopic to the identity map via the map $T : C_n(X) \to C_{n+1}(X)$ defined as follows.

Consider the map $\phi_\sigma : \Delta^n \times I \overset{\text{proj}}{\to} \Delta^n \overset{\sigma}{\to} X$ given by projection. Write $\sigma_*^\sigma : C_{n+1}(\Delta^n \times I) \to C_{n+1}(X)$ the induced map defined by $\sigma_*(\tau) = \phi_\sigma \circ \tau$. Then define

$$T(\sigma) = \sigma_*(T^\Delta(\Delta^n))$$

Proof We have that

$$\begin{aligned}
\partial S(\sigma) &= \partial(\sigma_*(S\Delta^n)) \\
&= \sigma_*(\partial(S\Delta^n)) && (\sigma_* \text{ is a chain map}) \\
&= \sigma_*(S(\partial\Delta^n)) && \text{(Lemma 4.2.4)} \\
&= \sum_{i=0}^n (-1)^i \sigma_*(S(\partial_i\Delta^n)) \\
&= \sum_{i=0}^n (-1)^i S(\partial_i\sigma) && (\text{Definition of } S) \\
&= S(\partial\sigma)
\end{aligned}$$

which shows that S is chain map.

Since T^Δ satisfies the chain homotopy relation and T and σ_* is linear, we have that T also satisfies the chain homotopy relation so that it is a chain homotopy from S to id. ■

We now come to the final ingredient of the proof. The subgroup of n-chains consists of linear combinations of n-chains contained in U_1 and U_2.

Definition 2.5.4 (Subgroup of n-Chains from Subspaces) Let X be a space and U_1, U_2 be open such that $X = U_1 \cup U_2$. Define

$$C_n(U_1 + U_2) = \left\{ \sum_{i \in I} \sigma_i + \sum_{j \in J} \tau_j \middle| \sigma_i \in C_n(U_1) \text{ and } \tau_i \in C_n(U_2) \right\}$$

the subgroup of $C_n(X)$ of n-chains that can be written as the sum of n-chains in U_1 and n-chains in U_2.

We can form a short exact sequence using the subgroup of n-chains. This allows to use the fact that short exact sequences produces a long exact sequence in homology groups.

Proposition 2.5.5 Let X be a space and U_1, U_2 be open such that $X = U_1 \cup U_2$. Let $j_1 : U_1 \to X$, $j_2 : U_2 \to X$ and $i_1 : U_1 \cap U_2 \to U_1$, $i_2 : U_1 \cap U_2 \to X$ be inclusions. Then the sequence of chain complexes

$$0 \longrightarrow C_\bullet(U_1 \cap U_2) \xrightarrow{(i_1)_* - (i_2)_*} C_\bullet(U_1) \oplus C_\bullet(U_2) \xrightarrow{(j_1)_* + (j_2)_*} C_\bullet(U_1 + U_2) \longrightarrow 0$$

is exact.

Proof We have to show exactness at the three spots in each degree n.

- Since the inclusion $(i_1)_* : C_n(U_1 \cap U_2) \to C_n(U_1)$ is already injective, so is $(i_1)_* - (i_2)_*$.
- We have that the composite map is equal to

$$((j_1)_* + (j_2)_*) \circ ((i_1)_* - (i_2)_*) = (j_1)_* \circ (i_1)_* - (j_2)_* \circ (i_2)_* = (j_1 \circ i_1)_* - (j_2 \circ i_2)_* = 0$$

since $j_1 \circ i_1$ and $j_2 \circ i_2$ are both inclusions from $U_1 \cap U_2$ to X. This shows that $\text{im}((i_1)_* - (i_2)_*) \subseteq \ker((j_1)_* + (j_2)_*)$.

Conversely, if $(c_1, c_2) \in C_n(U_1) \oplus C_n(U_2)$, such that $(j_1)_*(c_1) + (j_2)_*(c_2) = 0$, then we have that $(j_1)_*(c_1) = (j_2)_*(-c_2)$ so that c_1 and c_2 must both be in $U_1 \cap U_2$. This means that there exists some $c \in U_1 \cap U_2$ such that $k_*(c) = (j_1)_*(c_1) = (j_2)_*(-c_2)$ for some $k_* : U_1 \cap U_2 \to X$. By injectivity of $(i_1)_*$ and $(i_2)_*$, we deduce that $(i_1)_*(c) = c_1$ and $(i_2)_*(c) = c_2$.
- The subgroup $C_n(U_1 + U_2)$ is defined precisely as the image of $(j_1)_* + (j_2)_*$, so this map is surjective.

And so the sequence of chain complexes is exact. ∎

Note that the choice of the minus sign can be place arbitrarily. The important point is that the place of the minus is so that we can prove $\text{im}((i_1)_* - (i_2)_*) = \ker((j_1)_* + (j_2)_*)$.

Proposition 2.5.6 Let X be a space and U_1, U_2 be open such that $X = U_1 \cup U_2$. Then the inclusion

$$C_\bullet(U_1 + U_2) \hookrightarrow C_\bullet(X)$$

induces isomorphisms in homology.

Proof Let $\sigma \in C_n(X)$ be a singular n-simplex of X. Consider $\sigma^{-1}(U_1)$ and $\sigma^{-1}(U_2)$ as an open cover of Δ^n. Let δ be the Lebesgue number of the open cover. Suppose that $v_k \in S^k(\Delta^n)$ lie in the k-times barycentric subdivision of σ. By lemma 3.5.5, we have that

$$\text{diam}(v_k) \leq \left(\frac{n}{n+1} \right)^k \text{diam}(\Delta^n)$$

As $k \to \infty$, $\text{diam}(v_k) \to 0$. So there exists $k \in \mathbb{N}$ such that $\text{diam}(v_k) \leq \delta$. By definition of the Lebesgue cover this means that v_k lie completely in U_1 or U_2. Hence there exists a smallest number k such that $S^k(\sigma) \in C_n(U_1 + U_2)$.

Let $z \in Z_n(X)$ be an n-cycle in X. Thus z is a linear combination of finitely many n-simplexes $\sigma_k : \Delta^n \to X$. Choose $k \in \mathbb{N}$ such that $S^k(\sigma_i) \in C_n(U_1 + U_2)$. Since $S \simeq \text{id}$ by the proposition 4.3.3 and composition of chain homotopic maps are chain homotopic, we have that $S^k \simeq \text{id}$ via some chain homotopy η. Thus we have that

$$z - S^k(z) = \partial\eta(z) + \eta\partial(z) = \partial\eta(z)$$

since $\partial z = 0$. In particular, this shows that z and $S^k(z)$ are homologous in $H_n(X)$ so that $[z] = \iota([S^k(z)])$.

For injectivity, let $w \in Z_n(C_\bullet(U_1 + U_2))$ such that $\iota([w]) = 0$. This means that $w = \partial(z)$ for some $z \in C_{n+1}(X)$. Similar to the above, there exists $k \in \mathbb{N}$ such that $S^k(z) \in C_{n+1}(U_1 + U_2)$ and η such that $z - S^k(z) = \partial\eta(z) + \eta\partial(z)$. But then $\partial S^k(z) = \partial(z) - \partial^2\eta(z) - \partial\eta\partial(z) = w - \partial\eta(w)$ so that $[w] = 0$ as required. ■

We can now combine all the results to prove the Mayer-Vietoris sequence.

Theorem 2.5.7 (Mayer-Vietoris Sequence) Let $X = A \cup B$ be the union of two open subspaces with $j_1 : U_1 \to X$ and $j_2 : U_2 \to X$ the inclusion maps. Let $i_1 : A \cap B \to A$ and $i_2 : A \cap B \to B$ also be the inclusion maps. Then there exists connecting homomorphisms $\partial : H_n(X) \to H_{n-1}(U_1 \cap U_2)$ such that

$$\cdots \longrightarrow H_{n+1}(X) \xrightarrow{\partial} H_n(U_1 \cap U_2) \xrightarrow{(i_1)_* - (i_2)_*} H_n(U_1) \oplus H_n(U_2) \xrightarrow{(j_1)_* + (j_2)_*} H_n(X) \xrightarrow{\partial} H_{n-1}(U_1 \cap U_2) \longrightarrow \cdots$$

is a long exact sequence.

Proof The short exact sequence in proposition 4.3.6 induces a long exact sequence by theorem 1.4.2. By the above proposition, we can replace $H_n(C_\bullet(U_1 + U_2))$ by $H_n(X)$ and so we are done. ■

Note that the Mayer-Vietoris sequence also holds for reduced homology.

Our first application of the Mayer-Vietoris sequence comes with the computation of the homology groups of the k-spheres.

Example 2.5.8 Let $k \in \mathbb{N}$. Then the homology of the k-sphere S^k is

$$H_n(S^k) \cong \begin{cases} \mathbb{Z} & \text{if } n = k, 0 \\ 0 & \text{otherwise} \end{cases}$$

Proof We first consider the case of S^1. Since S^1 is path connected, $H_0(S^1) \cong \mathbb{Z}$. Moreover, $\pi_1(S^1, 1) \cong \mathbb{Z}$ is already abelian and so $H_1(X) \cong \mathbb{Z}$. Let U_1 be the upper half of S^1 and U_2 the lower half. It is clear that U_1 and U_2 are contractible, and that $U_1 \cap U_2 \simeq * \amalg *$. By Mayer Vietoris sequence and homotopy invariance, we have

$$\cdots \longrightarrow H_{n+1}(S^1) \longrightarrow H_n(* \amalg *) \longrightarrow H_n(*) \oplus H_n(*) \longrightarrow H_n(S^1) \longrightarrow \cdots$$

Combining the homology of one point space and the fact that homology can be decomposed into the homology of its path connected components, we have an exact sequence

$$0 \longrightarrow H_n(S^1) \longrightarrow 0$$

for all $n \geq 1$. This shows that

$$H_n(S^1) \cong \begin{cases} \mathbb{Z} & \text{if } n = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

and so we have computed the homology groups of S^1.

We now induct on k where S^k means the k-sphere in $\mathbb{R}^{k+1}$. Suppose that the homology of the sphere S^{k-1} is given by the formula. Write S^k as the union of the open upper hemisphere U_1 and the open lower hemisphere U_2 each containing the equator. Then $U_1 \cap U_2 \simeq S^{k-1}$ and U_1, U_2 are contractible. By Mayer Vietoris sequence and homotopy invariance, we have

$$\cdots \longrightarrow H_{n+1}(S^k) \longrightarrow H_n(S^{k-1}) \longrightarrow H_n(*) \oplus H_n(*) \longrightarrow H_n(S^k) \longrightarrow \cdots$$

S^k is path connected and so $H_0(S^k) = \mathbb{Z}$. We know that $\pi_1(S^k, x) = 0$ and thus $H_1(S^k) = 0$. Now consider the case of $n > 1$. Combining the homology of one point space and induction hypothesis, we have an exact sequence

$$0 \longrightarrow H_n(S^k) \longrightarrow H_{n-1}(S^{k-1}) \longrightarrow 0$$

Again using induction hypothesis, we see that

$$H_n(S^k) \cong H_{n-1}(S^{k-1}) \cong \begin{cases} \mathbb{Z} & \text{if } n = k \\ 0 & \text{otherwise} \end{cases}$$

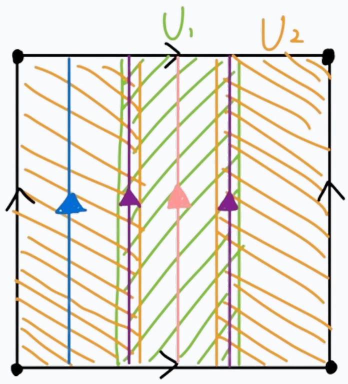
and so we conclude. ■

Recall that the torus and the Klein bottle can be expressed by a quotient of $I \times I$. Together with Mayer-Vietoris sequence and homotopy invariance we can compute the singular homology groups of the two.

Example 2.5.9 Let $\mathbb{T} = S^1 \times S^1$ denote the torus. Then the homology of the torus $\mathbb{T}$ is

$$H_k(\mathbb{T}) = \begin{cases} \mathbb{Z} & \text{if } k = 0, 2 \\ \mathbb{Z}^2 & \text{if } k = 1 \\ 0 & \text{otherwise} \end{cases}$$

Proof Cover the torus by two open sets.



The two open sets U_1, U_2 each deformation retract to a circle and $U_1 \cap U_2$ deformation retracts to a disjoint union of two circles. Using homotopy invariance, we obtain

$$H_k(U_1), H_k(U_2) \cong \begin{cases} \mathbb{Z} & \text{if } k = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

and since the disjoint union of two circles consists of two path connected components, together with proposition 3.1.7, we have that

$$H_k(U_1 \cap U_2) \cong \begin{cases} \mathbb{Z} \oplus \mathbb{Z} & \text{if } k = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

By the Mayer-Vietoris sequence, the non-trivial terms of the sequence are precisely given by

$$0 \longrightarrow \tilde{H}_2(T) \xrightarrow{\partial} \tilde{H}_1(U_1 \cap U_2) \xrightarrow{i} \tilde{H}_1(U_1) \oplus \tilde{H}_2(U_2) \xrightarrow{j} \tilde{H}_1(T) \xrightarrow{\partial} \tilde{H}_0(U_1 \cap U_2) \longrightarrow 0$$

where ∂ are the connecting homomorphisms, i and j are induced by the inclusion maps $i = (\iota_1)_* - (\iota_2)_*$ and $j = (j_1)_* + (j_2)_*$. Also because U_1, U_2 and T are all path connected the end terms are 0. Rewriting them into known homology groups, we obtain

$$0 \longrightarrow \tilde{H}_2(T) \xrightarrow{\partial} \mathbb{Z}^2 \xrightarrow{i} \mathbb{Z}^2 \xrightarrow{j} \tilde{H}_1(T) \xrightarrow{\partial} \mathbb{Z} \longrightarrow 0$$

Consider the sequence

$$0 \longrightarrow \tilde{H}_2(T) \xrightarrow{\partial} \mathbb{Z}^2 \xrightarrow{i} \text{im}(i) \longrightarrow 0$$

This sequence is exact by definition. Moreover, since $\text{im}(i)$ is a subgroup of the abelian group $\mathbb{Z}^2$, $\text{im}(i)$ is finite free and so is isomorphic to $\mathbb{Z}^n$ for some $0 \leq n \leq 2$. Then by proposition 1.2.6, the sequence is split exact and by proposition 1.2.5, we obtain that

$$\mathbb{Z}^2 \cong \tilde{H}_2(T) \oplus \text{im}(i)$$

Since $\text{im}(i)$ is a subgroup of $\mathbb{Z}^2$ we can divide both sides by $\text{im}(i)$ to obtain

$$\tilde{H}_2(T) \cong \frac{\mathbb{Z}^2}{\text{im}(i)} \cong \ker(i)$$

Also, by considering the split exact sequence

$$0 \longrightarrow \ker(\partial) \xrightarrow{\iota} \tilde{H}_1(T) \xrightarrow{\partial} \mathbb{Z} \longrightarrow 0$$

we obtain an isomorphism $\tilde{H}_1(T) \cong \mathbb{Z} \oplus \ker(\partial)$. Now $\ker(\partial) = \text{im}(j)$ since the sequence is exact. Also, we have that

$$\frac{\mathbb{Z}^2}{\ker(j)} \cong \text{im}(j)$$

by the first isomorphism theorem. Using the fact that the sequence being exact implies $\text{im}(i) = \ker(j)$, we have that

$$\tilde{H}_1(T) \cong \mathbb{Z} \oplus \text{im}(j) \cong \mathbb{Z} \oplus \frac{\mathbb{Z}^2}{\ker(j)} \cong \frac{\mathbb{Z}^2}{\text{im}(i)} \cong \mathbb{Z} \oplus \text{coker}(i)$$

We know have to compute what i does to the generators of $\tilde{H}_1(U_1 \cap U_2) \cong \mathbb{Z}$ marked in purple. Recall that i is defined as $(\iota_1)_* - (\iota_2)_*$. i then sends the generators to the positive generator (pink) in U_1. It sends the generators also to the positive generator (blue) in U_2. This means that our map i can be written as

$$i = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

with basis vectors the two generators in $\tilde{H}_1(U_1 \cap U_2)$ and the two in $\tilde{H}_1(U_1) \oplus \tilde{H}_2(U_2)$. The smith normal form of this map reduces it to the matrix

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

Its kernel is one copy of $\mathbb{Z}$ and its cokernel is also one copy of $\mathbb{Z}$. Thus we conclude that

$$H_k(\mathbb{T}) = \begin{cases} \mathbb{Z} & \text{if } k = 0, 2 \\ \mathbb{Z}^2 & \text{if } k = 1 \\ 0 & \text{otherwise} \end{cases}$$

and so we are done. ■

Note that the choice of the orientation of the generators is arbitrary, one can choose the opposite orientation and deduce the same homology groups.

Example 2.5.10 Let K denote the Klein bottle. Then the homology of the Klein bottle K is

$$H_k(K) = \begin{cases} \mathbb{Z} & \text{if } k = 0 \\ \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} & \text{if } k = 1 \\ 0 & \text{otherwise} \end{cases}$$

2.6 Three Applications of Singular Homology

In Algebraic Topology 1, we have seen Brouwer fixed point theorem for D the unit disc in $\mathbb{R}^2$. Using homology groups, we can finally state and prove the result for n-dimensional discs.

Lemma 2.6.1 For any $k \in \mathbb{N}$, S^{k-1} is not a retract of D^k.

Proof Assume to the contrary that $i : S^{k-1} \hookrightarrow D^k$ admits a retraction $r : D^k \to S^{k-1}$. Then $\text{id}_{S^k} = r \circ i$ so that

$$\widetilde{H}_{k-1}(S^{k-1}) \xrightarrow{i_*} \widetilde{H}_{k-1}(D^k) \xrightarrow{r_*} \widetilde{H}_{k-1}(S^{k-1})$$

is the identity map. Substituting the homology of S^{k-1} and D^k, we have that

$$\mathbb{Z} \xrightarrow{i_*} 0 \xrightarrow{r_*} \mathbb{Z}$$

is the identity map which is impossible. ∎

Theorem 2.6.2 (Brouwer Fixed-point Theorem) Every continuous map $f : D^k \to D^k$ has a fixed point.

Proof Suppose not, then the ray starting at $f(x)$ in the direction of x meets S^{k-1} in exactly one point $g(x) \neq f(x)$. Then $g : D^k \to S^{k-1}$ defines a retraction. By the above corollary, this is a contradiction. ∎

Using topological properties preserved by homeomorphisms, we are only able to prove that $\mathbb{R}^1$ and $\mathbb{R}^2$ are not homeomorphic using the argument on the number of connected components. Using homology, we can prove the full form of invariance of domain.

Corollary 2.6.3 (Invariance of Domain) If $n \neq m$, then $\mathbb{R}^n$ is not isomorphic $\mathbb{R}^m$.

Proof Assume that $\mathbb{R}^n \cong \mathbb{R}^m$. Then $f : \mathbb{R}^n \setminus \{0\} \to \mathbb{R}^m \setminus \{0\}$ is also a homeomorphism. Since $\mathbb{R}^n \setminus \{0\} \simeq S^{n-1}$ and $\mathbb{R}^m \setminus \{0\} \simeq S^{m-1}$, we have that f_* induces an isomorphism

$$\mathbb{Z} \cong \widetilde{H}_{n-1}(S^{n-1}) \cong \widetilde{H}_{n-1}(S^{m-1})$$

by homotopy invariance and the computation of the homology groups of the n-sphere. This can only be true when $n = m$ by our computations. ∎

The Jordan curve theorem is a highly non-trivial result that looks deceptively easy to prove. Using homology we can give an algebraic proof.

Definition 2.6.4 (Jordan Curve) A Jordan Curve is a simple closed curve in $\mathbb{R}^2$.

Lemma 2.6.5 Let $\gamma : I \to S^2$ be an injective continuous map with image $C = \gamma(I)$. Then

$$H_n(S^2 \setminus C) = \begin{cases} \mathbb{Z} & \text{if } n = 0 \\ 0 & \text{otherwise} \end{cases}$$

Proof Let $J \subseteq I$ be an interval. Define $C_J = \gamma(J)$. Let $U = S^2 \setminus C_{[0,1/2]}$ and $V = S^2 \setminus C_{[1/2,1]}$. Note that $U \cap V = S^2 \setminus C$ and

$$U \cup V = S^2 \setminus C_{[1/2,1/2]} \cong \mathbb{R}^2 \simeq *$$

By Mayer-Vietoris, we obtain isomorphisms

$$H_n(S^2 \setminus C) \cong H_n(S^2 \setminus C_{[0,1/2]}) \oplus H_n(S^2 \setminus C_{[1/2,1]})$$

for $n \geq 1$ together with a short exact sequence

$$0 \longrightarrow H_0(S^2 \setminus C) \longrightarrow H_0(S^2 \setminus C_{[0,1/2]}) \oplus H_0(S^2 \setminus C_{[1/2,1]}) \longrightarrow \mathbb{Z} \longrightarrow 0$$

Assume for a contradiction that for some $n \geq 1$ there exists $\sigma \in H_n(S^2 \setminus C)$ which is non-zero. By the isomorphism above, it remains non-zero in one of the two groups of the direct sum. By repeating this argument cutting the interval where its homology group contains σ in half, we obtain a nested sequence of intervals

$$I \supset J_1 \supset J_2 \supset \cdots$$

with non zero intersection $p \in \bigcap_{i=1}^{\infty} J_i$ such that $\sigma \neq 0$ in all of $H_n(S^2 \setminus C_{J_l})$. However we have that $S^2 \setminus C_{[p,p]} \cong \mathbb{R}^2$ is contractible hence σ vanishes in $H_n(S^2 \setminus \{\gamma(p)\})$. Let $\tau \in C_{n+1}(S^2 \setminus C_{[p,p]}$ such that $\partial\tau = \sigma$. Write τ as a finite linear combination of singular simplexes. Each of the singular simplexes has compact image in $S^2 \setminus C_{[p,p]}$ since Δ^{n+1} is compact. The union of these images is covered by open subsets $(S^2 \setminus C_{J_l})_l$ so by compactness, there exists l such that $\tau \in C_{n+1}(S^2 \setminus C_{J_l})$. But then $\sigma = 0$ in $H_n(S^2 \setminus C_{J_l})$, which is a contradiction. Thus $H_n(S^2 \setminus C) = 0$ for all $n \geq 1$.

For $n = 0$, assume that $x, y \in S^2 \setminus C$ are distinct in different path components. Then similar to the above we obtain a nested sequence of intervals J_l such that x and y are in different path components of $S^2 \setminus C_{J_l}$ for all l. Since $S^2 \setminus \{\gamma(p)\}$ is contractible, it must contain a path connecting x and y. By compactness, this path misses C_{J_l} for large l. Thus x and y lie in the same path component of $S^2 \setminus C_{J_l}$ for large l, which is a contradiction. We deduce that $H_0(S^2 \setminus C) = \mathbb{Z}$. ■

Theorem 2.6.6 (Jordan Curve Theorem) Let $\gamma : [0,1] \to \mathbb{R}^2$ be a Jordan curve with image C. Then $\mathbb{R}^2 \setminus C$ has two connected components.

Proof Choose S^1_+ and S^1_- the upper and lower hemicircles of S^1 so that the intersection is S^0. Define the following: $X = S^2 \setminus \gamma(S^0)$, $U_+ = S^2 \setminus \gamma(S^1_+)$, $U_- = S^2 \setminus \gamma(S^1_-)$ and $U_+ \cap U_- = S^2 \setminus C$. Since $S^1_\pm \cong I$, the homology groups of $U_\pm$ is given by the above lemma. Also $X \simeq S^1$. By Mayer-Vietoris applied to U, V, X, we obtain an exact sequence

$$0 \longrightarrow H_{n+1}(S^2 \setminus \gamma(S^0)) \longrightarrow H_n(S^2 \setminus C) \longrightarrow H_n(U_+) \oplus H_n(U_-)$$

for $n > 0$ in which the first and third term vanish so that $H_n(S^2 \setminus C) = 0$. For $n = 0$, we have an exact sequence

$$0 \longrightarrow H_1(S^2 \setminus \gamma(S^0)) \longrightarrow \widetilde{H}_0(S^2 \setminus C) \longrightarrow \widetilde{H}_0(U_+) \oplus \widetilde{H}_0(U_-)$$

where the last term vanishes. Thus we have an isomorphism $\widetilde{H}_0(S^2 \setminus C) \cong \mathbb{Z}$. It follows that $H_0(S^2 \setminus C) \cong \mathbb{Z}^2$. So now we have

$$H_n(S^2 \setminus C) = \begin{cases} \mathbb{Z}^2 & \text{if } n = 0 \\ 0 & \text{otherwise} \end{cases}$$

Now consider $X = S^2 \setminus C$ and $U = \mathbb{R}^2 \setminus C$ identified in $\mathbb{R}^2 \cup \{\infty\} \cong S^2$. Let V be an open disk around ∞ that does not meet C. Using Mayer-Vietoris sequence, the only interesting terms are this sequence

$$0 \longrightarrow H_n(\mathbb{R}^2 \setminus C) \longrightarrow 0$$

for $n \geq 2$, which gives $H_n(\mathbb{R}^2 \setminus C) = 0$ and the lower degree terms give an exact sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow H_1(\mathbb{R}^2 \setminus C) \longrightarrow 0 \longrightarrow 0 \longrightarrow \widetilde{H}_0(\mathbb{R}^2 \setminus C) \longrightarrow \mathbb{Z} \longrightarrow 0$$

which is due to the fact that $U \cap V \simeq S^1$ and V is contractible. Then it is clear that $H_1(\mathbb{R}^2 \setminus C) \cong \mathbb{Z}$ and $\widetilde{H}_0(\mathbb{R}^2 \setminus C) \cong \mathbb{Z}$ so that $H_0(\mathbb{R}^2 \setminus C) \cong \mathbb{Z}^2$ and so we obtain

$$H_n(\mathbb{R}^2 \setminus C) = \begin{cases} \mathbb{Z}^2 & \text{if } n = 0 \\ \mathbb{Z} & \text{if } n = 1 \\ 0 & \text{if } n > 1 \end{cases}$$

Since $\mathbb{R}^2 \setminus C$ is locally path connected, by corollary 3.2.3 we thus have that $\mathbb{R}^2 \setminus C$ has two connected components. ■

3 Relative Homology

3.1 Relative Homology Groups

Given a subspace A of a space X, we know that the inclusion map $A \hookrightarrow X$ induces an inclusion $C_n(A) \hookrightarrow C_n(X)$. Unfortunately, this does not induce an injection $H_n(A) \to H_n(X)$. Relative homology gives a precise measure of the failure of injectivity and surjectivity of the map in homology.

Definition 3.1.1 (Relative Homology Group) Let X be a topological space and $A \subseteq X$ a subspace. Define the relative homology group $H_n(X, A)$ to be the homology group of the chain complex

$$\cdots \longrightarrow C_n(X, A) \xrightarrow{\partial} C_{n-1}(X, A) \longrightarrow \cdots$$

where $C_n(X, A)$ denotes the quotient group $C_n(X)/C_n(A)$. In other words,

$$H_n(X, A) = \frac{\ker(\partial : C_n(X, A) \to C_{n-1}(X, A))}{\operatorname{im}(\partial : C_{n+1}(X, A) \to C_n(X, A))} = H_n(C_\bullet(X, A))$$

Elements of $Z_n(X, A)$ are called relative n-cycles, while elements of $B_n(X, A)$ are called relative n-boundaries.

Geometrically, relative n-cycles are n-cycles in $C_n(X)$ such that $\partial z \in C_{n-1}(A)$ which means that the boundary of z is contained in the subspace A. Intuitively, we are treating cycles in the subspace A as 0 and so the homology groups measure the homology of X without A.

TBA: Functoriality

Relative homology also satisfies the homotopy invariance property and has the Mayer-Vietoris sequence.

Theorem 3.1.2 (Homotopy Invariance) Let $(X, A), (Y, B)$ be pairs of spaces. Let $f, g : (X, A) \to (Y, B)$ be homotopic maps. Then the induced map on relative homology are the same:

$$f_* = g_* : H_n(X, A) \to H_n(Y, B)$$

Corollary 3.1.3 Let $f : (X, A) \to (Y, B)$ be a map such that both $f : X \to Y$ and $f|_A : A \to B$ are homotopy equivalences. Then

$$f_* : H_n(X, A) \to H_n(Y, B)$$

induces an isomorphism for all $n \in \mathbb{N}$.

Definition 3.1.4 (Triplet of Spaces) Let X be a space. A triple (X, A, B) is said to be a triplet of spaces if $B \subseteq A \subseteq X$.

Definition 3.1.5 (Maps of Triplets) Let (X, A, B) and (Y, C, D) be triplets of spaces. Let $f; X \to Y$ be a map. We say that f is a map of the triplets if $f(A) \subseteq C$ and $f(B) \subseteq D$.

Proposition 3.1.6 Let (X, A, B) be a triplet of spaces. Let $j : B \hookrightarrow A$ and $i : A \hookrightarrow X$ be inclusions. Then

$$0 \longrightarrow C_\bullet(A, B) \xrightarrow{i} C_\bullet(X, B) \xrightarrow{j} C_\bullet(X, A) \longrightarrow 0$$

is a short exact sequence of chain complexes.

TBA: Naturality

Proposition 3.1.7 (Long Exact Sequence of a Triple) Let (X, A, B) be a triplet of spaces. Let $j : B \hookrightarrow A$ and $i : A \hookrightarrow X$ be inclusions. Then there is a long exact sequence

$$\cdots \longrightarrow H_n(A, B) \xrightarrow{i_*} H_n(X, B) \xrightarrow{j_*} H_n(X, A) \xrightarrow{\partial} H_{n-1}(A, B) \longrightarrow \cdots$$

in relative homology induced by the inclusion maps and the connecting homomorphism ∂ from the snake lemma.

Theorem 3.1.8 (Relative Mayer-Vietoris Sequence) Let (X, Y) be a pair of space such that $A, B \subset X$ cover X and $C, D \subset Y$ cover Y with $C \subset A$ and $D \subset B$. Then there is an exact sequence

$$\cdots \longrightarrow H_n(A \cap B, C \cap D) \xrightarrow{\quad\Phi\quad} H_n(A, C) \oplus H_n(B, D) \xrightarrow{\quad\Psi\quad} H_n(X, Y) \longrightarrow \cdots$$

in relative homology.

3.2 Relation to Singular Homology

Theorem 3.2.1 (Exact Sequence of Pairs of Spaces) Let (X, A) be a pair of spaces. Then there is an exact sequence

$$\cdots \longrightarrow H_n(A) \xrightarrow{\iota_*} H_n(X) \xrightarrow{j_*} H_n(X, A) \xrightarrow{\partial} H_{n-1}(A) \longrightarrow \cdots$$

where
- $\iota : A \to X$ is the inclusion map
- $j : X \to X \setminus A$ is the quotient map
- The connecting homomorphism $\partial : H_n(X, A) \to H_{n-1}(A)$ is defined by $[z] \mapsto [dz]$ for $z \in C_n(X)$ a relative cycle.

Moreover, the exact sequence is natural with respect to maps $f : (X, A) \to (Y, B)$.

Proof Notice that

$$0 \longrightarrow C_\bullet(A) \xrightarrow{\iota} C_\bullet(X) \xrightarrow{j} C_\bullet(X, A) \longrightarrow 0$$

is a short exact sequence by construction. Thus it induces a long exact sequence in homology groups.

Consider $[z] = z + C_n(A) \in C_n(X, A)$ a cycle in $C_n(X, A)$. The surjective map j is the projection map so $j(z) = [z]$. Now $d(z) \in \ker(j)$ because

$$j(d(z)) = d(j(z)) = d([z]) = 0$$

by assumption So $d(z) \in \ker(j) = \operatorname{im}(i)$. Since i is the inclusion, $[d(z)] \in C_{n-1}(A)$ is precisely the element that ∂ maps $[z]$ to. ■

We obtain an immediate result of the singular homology of the space X and its subspace A.

Lemma 3.2.2 Let (X, A) be a pair of spaces. The inclusion map $\iota : A \to X$ induces an isomorphism

$$H_n(A) \cong H_n(X)$$

for all $n \in \mathbb{N}$ if and only if $H_n(X, A) = 0$ for all n.

Proof If $H_n(X, A) = 0$ for all $n \in \mathbb{N}$, then the long exact sequence above shows that there are isomorphisms $H_n(A) \cong H_n(X)$. If $H_n(A) \cong H_n(X)$ in the long exact sequence above, then the map $H_n(X) \to H_n(X, A)$ is the zero map and the map $H_n(X, A) \to H_{n-1}(A)$ is the zero map. Thus for $n \geq 0$ there is an exact sequence

$$0 \longrightarrow H_n(X, A) \longrightarrow 0$$

showing that $H_n(X, A) = 0$. ■

Using the lemma, we have a low degree interpretation for relative homology.

Proposition 3.2.3 Let (X, A) be a pair of space. Then the following are true.
- $H_0(X, A) = 0$ if and only if every path component of X contains at least one path component of A.
- $H_1(X, A) = 0$ if and only if $H_1(A) \to H_1(X)$ is surjective and each path component of X contains at most one path-component of A.

Lemma 3.2.4 Let X be space and $x \in X$ be a point. Then there is an isomorphism

$$H_n(X, \{x\}) \cong \widetilde{H}_n(X)$$

of homology groups for all $n \in \mathbb{N}$.

Proof We have a long exact sequence as in theorem 4.1.3. In particular, since $H_n(\{x\}) = 0$ for all $n \geq 1$, we have an isomorphism

$$H_n(X, \{x\}) \cong H_n(X)$$

which descends to an isomorphism in reduced homology. We are now left with the exact sequence:

$$0 \longrightarrow H_1(X) \xrightarrow{j_*} H_1(X, x) \xrightarrow{\partial} H_0(x) \xrightarrow{\iota_*} H_0(X) \xrightarrow{j_*} H_0(X, x) \longrightarrow 0$$

Now since $\iota : \{x\} \to X$ is the inclusion map, the projection map $p : X \to \{x\}$ is such that $p \circ \iota = \text{id}$. By proposition 3.1.6 we have $p_* \circ \iota_\circ = \text{id}_*$ and thus ι_* is injective and has kernel 0. Exactness then implies that $\text{im}(\partial) = \ker(\iota_*) = 0$ and thus ∂ is the zero map. This means that $\widetilde{H}_1(X) \cong H_1(X) \cong H_1(X, x)$.

We are now left with a short exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\iota_*} H_0(X) \xrightarrow{j_*} H_0(X, x) \longrightarrow 0$$

Since $H_0(*) \cong \mathbb{Z}$. The map $p_* : H_0(X) \to H_0(x)$ has the property that $p_* \circ \iota_* = \text{id}_*$. This means that the short exact sequence is split exact and we have that

$$H_0(X) \cong \mathbb{Z} \oplus H_0(X, x)$$

Consider the map

$$H_0(X) \underset{\cong}{\xrightarrow{p_* \times j_*}} H_0(x) \oplus H_0(X, x) \xrightarrow{\pi} H_0(x)$$

where $\pi : H_0(x) \oplus H_0(X, x) \to H_0(x)$ is defined by $\pi(a, c) = a$. This map sends $b \in H_0(X)$ to $p_*(b)$. Then we have

$$\begin{aligned}
\widetilde{H}_0(X) &\cong \ker(H_0(X) \to H_0(x)) \\
&\cong \ker(H_0(X) \oplus H_0(X, x) \stackrel{p_*}{\to} H_0(x)) \\
&= H_0(X, x)
\end{aligned}$$

Therefore we conclude. ■

3.3 The Excision Theorem

The excision theorem is a powerful theorem for computing relative homology groups. This statement is derived directly from Mayer-Vietoris.

Theorem 3.3.1 (The Excision Theorem) Let X be a space and Z, A be subspaces of X such that $\overline{Z} \subseteq A^\circ$. Then the inclusion map $(X \setminus Z, A \setminus Z) \hookrightarrow (X, A)$ induces an isomorphism

$$H_n(X \setminus Z, A \setminus Z) \cong H_n(X, A)$$

for all n. Moreover, the excision isomorphism is natural in maps $f : (X, A, Z) \to (Y, B, C)$.

Proof Let $B = X \setminus Z$. Then notice that $A \cap B = A \setminus Z$, $A^\circ \cup B^\circ = A^\circ \cup (X \setminus \overline{Z}) = X$. Moreover, we have that

$$\begin{aligned} C_n(X \setminus Z, A \setminus Z) &= C_n(B, A \cap B) \\ &= \frac{C_n(B)}{C_n(A \cap B)} && \text{(By definition)} \\ &\cong \frac{C_n(A+B)}{C_n(A)} && \text{(Second Isomorphism Theorem)} \end{aligned}$$

This implies that

$$0 \longrightarrow C_\bullet(A) \longrightarrow C_\bullet(A+B) \longrightarrow C_\bullet(X \setminus Z, A \setminus Z) \longrightarrow 0$$

is a short exact sequence of chain complexes. Moreover, by considering inclusion maps, we have the following commutative diagram:

$$\begin{tikzcd} 0 \arrow[r] & C_\bullet(A) \arrow[r] \arrow[d, "="] & C_\bullet(A+B) \arrow[r] \arrow[d] & C_\bullet(X \setminus Z, A \setminus Z) \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & C_\bullet(A) \arrow[r] & C_\bullet(X) \arrow[r] & C_\bullet(X, A) \arrow[r] & 0 \end{tikzcd}$$

Considering that the rows are short exact sequences of chain complexes, they induce long exact sequences in homology by naturality in theorem 1.4.3:

$$\begin{tikzcd} \cdots \arrow[r] & H_n(A) \arrow[r] \arrow[d, "="] & H_n(A+B) \arrow[r] \arrow[d] & H_n(X \setminus Z, A \setminus Z) \arrow[r] \arrow[d] & H_{n-1}(A) \arrow[r] \arrow[d, "="] & H_{n-1}(A+B) \arrow[r] \arrow[d] & \cdots \\ \cdots \arrow[r] & H_n(A) \arrow[r] & H_n(X) \arrow[r] & H_n(X, A) \arrow[r] & H_{n-1}(A) \arrow[r] & H_{n-1}(X) \arrow[r] & \cdots \end{tikzcd}$$

where the vertical arrows are naturally induced by inclusion. By proposition 8.2.3, the second the fifth arrows are isomorphisms. Thus by the five lemma, we have that

$$H_n(X \setminus Z, A \setminus Z) \cong H_n(X, A)$$

and so we are done. ■

Lemma 3.3.2 The Excision Theorem is equivalent to the following version: If $A, B \subset X$ are subspaces such that $X = A^\circ \cup B^\circ$, then the inclusion $(B, A \cap B) \to (X, A)$ induces isomorphisms

$$H_n(B, A \cap B) \xrightarrow{\cong} H_n(X, A)$$

for all n.

When X and A in relative homology is nice enough, we can interpret the result from relative homology as the homology groups of the quotient space. The following notation is due to Hatcher.

Definition 3.3.3 (Good Pairs) Let X be a space and A a closed subspace of X. We say that the pair (X, A) is a good pair if there exists an open set V such that $A \subset V$ and V deformation retracts to A.

It is clear by definition that any CW-complex X and subcomplex A forms a good pair.

Lemma 3.3.4 Both CW-complexes and Δ-complexes are good pairs.

Proposition 3.3.5 Let X be a space and A a closed subspace of X such that (X, A) is a good pair. Then the quotient map $X \to X/A$ induces a natural isomorphism

$$H_n(X, A) \cong \widetilde{H}_n(X/A)$$

in homology groups together with the excision isomorphism.

Proof Let V be a open neighbourhood of A satisfying the good pair condition. Then there is an obvious inclusion map $(X, A) \to (X, V)$. Applying the long exact sequence of relative homology for both (X, A) and (X, V), we obtain the following diagram by naturality in theorem 1.4.3:

$$\begin{tikzcd} \cdots \arrow[r] & H_n(A) \arrow[r] \arrow[d, "\cong"] & H_n(X) \arrow[r] \arrow[d, "\cong"] & H_n(X,A) \arrow[r] \arrow[d] & H_{n-1}(A) \arrow[r] \arrow[d, "\cong"] & H_{n-1}(X) \arrow[r] \arrow[d, "\cong"] & \cdots \\ \cdots \arrow[r] & H_n(V) \arrow[r] & H_n(X) \arrow[r] & H_n(X,V) \arrow[r] & H_{n-1}(V) \arrow[r] & H_{n-1}(X) \arrow[r] & \cdots \end{tikzcd}$$

where the isomorphisms come from the fact that V deformation retracts onto V since (X, A) is a good pair. By the five lemma, we obtain an isomorphism

$$H_n(X, A) \cong H_n(X, V)$$

By excision, we obtain another isomorphism

$$H_n(X, A) \cong H_n(X, V) \cong H_n(X \setminus A, V \setminus A)$$

Repeating the argument with $(X/A, A/A)$, we obtain an isomorphism

$$H_n(X/A, A/A) \cong H_n(X \setminus A, V \setminus A) \cong H_n\left(\left(\frac{X}{A}\right) \setminus \left(\frac{A}{A}\right), \left(\frac{V}{A}\right) \setminus \left(\frac{A}{A}\right)\right)$$

Combining the two gives the desired result. ■

Most pairs of spaces are good pairs. We provide some examples that are not.

The Hawaiian earrings with its wedge point is not a good pair since any open neighbourhood of the wedge point contains an infinite number of circles and so cannot be contractible.

The interval together with $A = \left\{\frac{1}{n} \mid n \in \mathbb{N} \setminus \{0\}\right\}$ is not a good pair. In fact, one can show that the pair $(X/A, A/A)$ is homeomorphic to the Hawaiian earring with the wedge point.

Proposition 3.3.6 Let $k \geq 0$, we have that

$$H_n(\Delta^k, \partial\Delta^k) = \begin{cases} \mathbb{Z} & \text{if } n = k \\ 0 & \text{otherwise} \end{cases}$$

Proof Since Δ^k is also a CW-complex, $(\Delta^k, \partial\Delta^k)$ is a good pair thus we can apply proposition 6.2.4 to obtain

$$\widetilde{H}_n(\Delta^k, \partial\Delta^k) \cong \widetilde{H}_n(S^k)$$

and so the desired results follows. ■

Lemma 3.3.7 Assume that each (X_i, x_i) is a good pair. Then we have an isomorphism

$$\widetilde{H}_n\left(\bigvee_{i \in I}(X_i, x_i)\right) = \bigoplus_{i \in I} \widetilde{H}_n(X_i)$$

in reduced homology.

3.4 Local Homology Groups

Local homology groups forgets every homological information about the space X, under than a small neighbourhood around some subset $A \subseteq X$.

Definition 3.4.1 (The Local Homology Groups) Let X be a space. Let $A \subseteq X$ be a subspace. Define the local homology group of X in A to be the homology groups
$$H_n(X \mid A) = H_n(X, X \setminus A)$$
for each $n \in \mathbb{N}$.

Elements of the local homology group are represented by cycles in X whose boundary lie outside of x. It is called local because it only captures local topological data of X surrounding x.

Proposition 3.4.2 For all $n, k \in \mathbb{N}$, there is an isomorphism
$$H_n(\mathbb{R}^k \mid \{*\}) \cong \begin{cases} \mathbb{Z} & \text{if } n = k \\ 0 & \text{if } n \neq k \end{cases}$$

Proof Now if $k = 0$, the results are clear. If $k \geq 1$, then the long exact sequence of the pair $(\mathbb{R}^k, \mathbb{R}^k \setminus *)$ together with the fact that $\mathbb{R}^k \setminus * \simeq S^{k-1}$ and $\mathbb{R}^k \simeq *$ gives
$$H_n(\mathbb{R}^k \mid *) = 0$$
for $n > k$ and $n < k$. When $n = k$, we have an exact sequence
$$0 \longrightarrow H_k(\mathbb{R}^k \mid *) \longrightarrow H_{k-1}(S^{k-1}) \longrightarrow H_{k-1}(\mathbb{R}^k)$$
when $k > 1$ since $H_{k-1}(\mathbb{R}^k) = 0$. Thus $H_k(\mathbb{R}^k \mid *) \cong \mathbb{Z}$. If $k = 1$, then the last map $H_0(S^0) \to H_0(\mathbb{R})$ is given by the matrix $\begin{pmatrix} 1 & 1 \end{pmatrix} : \mathbb{Z}^2 \to \mathbb{Z}$ thus also giving isomorphism. ∎

The excision theorem gives the following stronger version of invariance of domain.

Corollary 3.4.3 Let $U \subseteq \mathbb{R}^m$ and $V \subseteq \mathbb{R}^n$ be non empty open subsets. If $U \cong V$ then $m = n$.

Proof Let $x \in U$. By excision, we obtain an isomorphism
$$H_k(U \mid \{x\}) \cong H_k(\mathbb{R}^m \mid \{x\}) \cong \begin{cases} \mathbb{Z} & \text{if } m = k \\ 0 & \text{if } n \neq k \end{cases}$$
and similarly for $H_k(V \mid \{\phi(x)\})$ if $\phi : U \to V$ gives the homeomorphism. Then it is clear that if the homology groups are equal, then $m = n$. ∎

This shows that no connected manifold can have atlases mapping to different dimensions of $\mathbb{R}$ so that the notion of dimension is well defined.

4 Degree Theory and Cellular Homology

4.1 Degree of Continuous Maps

The degree is a main component of the explicit construction of maps in cellular homology. Therefore we develop some of the basic theory here. Degree theory is also an important subject in its own right.

Definition 4.1.1 (Degree of a Continuous Map) Let $f : S^k \to S^k$ be a continuous map. Let

$$f_* : \widetilde{H}_k(S^k) \cong \mathbb{Z} \to \widetilde{H}_k(S^k) \cong \mathbb{Z}$$

be induced by f and let $f_*(1) = n$. Define the degree of f to be $\deg(f) = n$.

Lemma 4.1.2 Let $f, g : S^k \to S^k$ be continuous maps. Then the following are true regarding the degree.

- $\deg(\text{id}) = 1$
- $\deg(g \circ f) = \deg(g) \cdot \deg(f)$
- If $f \simeq g$ then $\deg(g) = \deg(f)$
- If f is a homotopy equivalence then $\deg(f) = \pm 1$
- If f is not surjective then $\deg(f) = 0$.

Proof

- The identity map on the k-sphere induces the identity map on the homology groups.
- Direct consequence of functoriality of the induced map.
- Direct consequence of homotopy invariance.
- If f is a homotopy equivalence that $f \simeq$ id. By the third and first point we are done.
- Let $x \in S^k$ not be in the image of f. Then we obtain a factorization of f.

$$\begin{tikzcd} S^k \arrow[rr, "f"] \arrow[dr] & & S^k \\ & S^k \setminus \{x\} \arrow[ur] & \end{tikzcd}$$

  Passing to reduced homology, we see that f_* factors through 0:

$$\begin{tikzcd} \mathbb{Z} \arrow[rr, "f_*"] \arrow[dr] & & \mathbb{Z} \\ & 0 \arrow[ur] & \end{tikzcd}$$

  so that f_* is the 0 map.

and so we conclude. ■

It is easy to compute all the possible degrees when $k = 0$ or 1. When $k = 0$, there are two points to map to and so there are two obvious maps: the identity and the continuous map taking one point to the other. Any other map must be homotopic to one of these two thus the only possible degree maps of S^0 is just the identity with degree 0 and the non-trivial map with degree 1.

When $k = 1$, the set of maps $f : S^1 \to S^1$ defined by $z \mapsto z^n$ has degree n. Indeed from Algebraic Topology 1 we have seen that every $\omega_n : I \to S^1$ defined by $t \mapsto e^{2\pi int}$ induces a map of sending the generator in $H_1(S^1) \cong \pi_1(S^1, 1)$ to n times the generator. Explicitly, if $\sigma : I \to S^1$ is the generator in $H_1(S^1)$ defined by $\sigma(t) = e^{2\pi it}$, then $f \circ \sigma$ is homologous to $n\sigma$ again via $\pi_1(S^1, 1) \cong H_1(S^1)$.

Proposition 4.1.3 For all $k \geq 1$ and for all $n \in \mathbb{Z}$, there exists a continuous map $f : S^k \to S^k$ of degree n.

Proof Define $SX = \frac{X\times[-1,1]}{X\times\{-1\},X\times\{1\}}$ (the suspension of X). The two open sets

$$C_+X = \frac{X\times(\epsilon,1]}{X\times\{1\}} \quad \text{and} \quad C_-X = \frac{X\times[-1,\epsilon)}{X\times\{-1\}}$$

are contractible. By Mayer-Vietoris sequence, we obtain an isomorphism

$$H_{k+1}(SX) \overset{\partial}{\cong} H_k(X)$$

It is clear that every map $f : X \to Y$ induces a map $Sf : SX \to SY$ by $Sf(x,t) = (f(x),t)$. Now notice that the following diagram

$$\begin{tikzcd}
H_{k+1}(SX) \arrow[r, "Sf_*"] \arrow[d, "\partial"'] & H_{k+1}(SY) \arrow[d, "\partial"] \\
H_k(X) \arrow[r, "f_*"'] & H_k(Y)
\end{tikzcd}$$

commutes by naturality in theorem 1.4.3. Applying this with $X = Y = S^{k-1}$ and since $S(S^{k-1}) \cong S^k$, we deduce that $\deg(Sf) = \deg(f)$ so if we use induction, we are done. But the base case is already treated by the above paragraph, and so we conclude. ■

Using the fundamental class of a sphere, we can compute the degree of a reflection.

Lemma 4.1.4 Let $S^k \subseteq \mathbb{R}^{k+1}$ be the unit circle. Let $f : S^k \to S^k$ be the reflection of a hyperplane through 0 in $\mathbb{R}^{k+1}$. Then $\deg(f) = -1$.

Proof The top S^k_+ and bottom S^k_- half of the sphere cut by the hyperplane are homeomorphic to k-simplicies. Choose k-simplicies $\sigma_+ : \Delta^k \xrightarrow{\cong} S^k_+$ and $\sigma^k_- : \Delta^k \xrightarrow{\cong} S^k_-$ such that $\partial\Delta^k$ is mapped homeomorphically to the equator $S^k_+ \cap S^k_-$ in both cases, and that the composition

$$\partial\Delta^k \xrightarrow{\sigma_+} S^k_+ \cap S^k_- \xrightarrow{(\sigma_-)^{-1}} \partial\Delta^k$$

is the identity.

It is clear that it is a cycle since it is a boundary in the chain complex $C_\bullet(\Delta^{k+1})$. I claim that $[\sigma_+ - \sigma_-]$ is a generator of $H_k(S^k)$.

We proceed by induction. When $k = 0$, the statement is clear. So suppose that $k > 0$. Let U_1, U_2 be open subspaces of Δ^{k+1} as follows. U_1 is an open neighbourhood of the last face of $\partial_{k+1}\Delta^{k+1}$ which deformation retracts onto $\partial\Delta^{k+1}$. U_2 is an open neighbourhood of the remaining faces $U_2 = \bigcup_{i=0}^k \partial_i\Delta^{k+1}$ which deformation retracts onto $\bigcup_{i=0}^k \partial_i\Delta^{k+1}$. Moreover, choose them in such a way that $U_1 \cap U_2$ deformation retract onto $\partial\partial\Delta^{k+1} = \partial[v_0,\dots,v_k]$ and $U_1 \cup U_2$ deformation retracts onto $\partial[v_0,\dots,v_{k+1}]$. By induction hypothesis, we know that $\widetilde{H}_{k-1}(U_1\cap U_2) \cong \mathbb{Z}$ is generated by

$$\partial([v_0,\dots,v_k]) = \sum_{i=0}^k (-1)^i [v_0,\dots,\hat{v}_i,\dots,v_k]$$

From Mayer-Vietoris sequence, since $U_1 \cup U_2$ deformation retracts onto $\partial\Delta^{k+1}$, the connecting homomorphism

$$\widetilde{H}_k(U_1\cup U_2) \to \widetilde{H}_{k-1}(U_1\cap U_2)$$

since U_1 and U_2 are contractible so we only need to show that $\partial\sigma$ is sent to the generator or its negative.

For this we will explicitly compute the connecting homomorphism. It is clear that

$$\left((-1)^{k+1}[v_0,\dots,v_k],\sum_{i=0}^{k}(-1)^i[v_0,\dots,\hat{v}_i,\dots,v_{k+1}]\right)\in C_k(U_1)\oplus C_k(U_2)$$

is such that it is a lift of the cycle $\partial\sigma$. Its image under the connecting homomorphism is the unique $(k-1)$-cycle τ in $U_1\cap U_2$ which satisfies

$$((i_1)_*(\tau),-(i_2)_*(\tau))=\left((-1)^{k+1}\partial([v_0,\dots,v_k]),\sum_{i=0}^{k}(-1)^i\partial([v_0,\dots,\hat{v}_i,\dots,v_{k+1}])\right)$$

It is clear that $\tau=(-1)^{k+1}\partial([v_0,\dots,v_k])$ is a generator in $\widetilde{H}_k(U_1\cap U_2)$.

We have seen that $s=[\sigma_+-\sigma_-]$ generates $H_k(S^k)$. Thus

$$f_*(s)=[f\circ\sigma_+]-[f\circ\sigma_-]=[\sigma_-]-[\sigma_+]=-f_*(s)$$

and thus the degree of f is -1 since it send a the generator to the negative of itself. ■

Proposition 4.1.5 Let $T:\mathbb{R}^{k+1}\to\mathbb{R}^{k+1}$ be a linear orthogonal transformation. The restriction of T to the homeomorphism

$$f:S^k\to S^k$$

has degree $\deg(f)=\det(T)$.

Proof Choose an orthonormal basis for $\mathbb{R}^{k+1}$ such that T can be represented by a block sum matrix where each block is either a 2×2 rotation or a 1×1 matrix with entry ± 1. This is possible from the notes Geometry. Any rotation is homotopic to the identity by undoing the rotation be rewinding. Thus T is homotopic to a diagonal matrix with m entries having -1 and $k+1-m$ entries having 1. But then T is just the composition of m reflections and $\det(T)=(-1)^m$. By the above lemma, we conclude. ■

As a result, we can compute the degree of the following two maps.

Corollary 4.1.6 Let $f:S^k\to S^k$ be the antipodal map. Then $\deg(f)=(-1)^{k+1}$.

Proof The antipodal map is a composition of $n+1$ reflections. By the above proposition we conclude. ■

Corollary 4.1.7 Let $f:S^k\to S^k$ have no fixed points. Then $\deg(f)=(-1)^{k+1}$.

Proof The line through $f(x)$ and $-x$ passes through the origin if and only if $f(x)=x$. Since f has no fixed points, the line never passes through the origin so that the map $g:S^k\times I\to S^k$ defined by

$$g(x,t)=\frac{tf(x)+(1-t)(-x)}{\|tf(x)+(1-t)(-x)\|}$$

defines a homotopy from f to the antipodal map. By homotopy invariance and the above corollary, we conclude that $\deg(f)=(-1)^{k+1}$. ■

We can now prove a famous result from manifold theory. Recall that a tangent vector field $v:M\to TM$ on a smooth manifold is a vector field such that $v(x)$ and x is orthogonal.

Theorem 4.1.8 (Hairy Ball Theorem) Every tangent vector field on an even-dimensional sphere vanishes at some point.

Proof Assume the contrary. This means that there exists a vector field $v : S^k \to \mathbb{R}^{k+1}$ that is non-zero everywhere, where k is even. Consider the map $g : S^k \times I \to \mathbb{R}^{k+1}$ defined by

$$g(x,t) = \cos(\pi t)x + \sin(\pi t)v(x)$$

Since x and $v(x)$ are orthogonal and $v(x) \neq 0$, the two vectors x and $v(x)$ are linearly independent. It follows that the map lands in $\mathbb{R}^{k+1} \setminus \{0\}$ and we can divide by the norm to obtain a homotopy $S^k \times I \to S^k$. This homotopy is in fact a homotopy from the identity to the antipodal map. They have degree 1 and -1 respectively. But by homotopy invariance of the degree, it is a contraction. ■

4.2 Local Degree

It is not at all obvious that given a map of spheres, how one would compute the degree of the map. We can do this by computing locally what the degree of the map looks like. These are called the local degree of f.

Definition 4.2.1 (Local Degree) Let $f : S^k \to S^k$ be a continuous map. Assume that $f^{-1}(y) = \{x_1, \dots, x_n\}$. Let U_i be an open neighbourhood of x_i such that $U_i \cap U_j = \emptyset$ for each $i \neq j$. Define the local degree of f to be the degree of the map

$$f|_{x_i} : H_k(U_i \mid x_i) \to H_k(S^k \mid y)$$

denoted by $\deg(f|_{x_i})$.

Notice that this indeed makes sense. Indeed by excision, we have an isomorphism

$$H_k(S^k, S^k \setminus \{x_i\}) \cong H_k(U_i, U_i \setminus \{x_i\})$$

by excising the piece $S^k \setminus U$. By using excision again, we obtain an isomorphism

$$H_k(S^k, S^k \setminus \{x_i\}) \cong H_k(\mathbb{R}^k, \mathbb{R}^k \setminus *)$$

Indeed S^k is a smooth manifold ans so there exists some $V \subseteq S^k$ which maps homeomorphically to an open subset of $\mathbb{R}^k$. Excision then gives

$$H_k\left(S^k \setminus (S^k \setminus V), (S^k \setminus \{x\}) \setminus (S^k \setminus V)\right) \cong H_k(S^k, S^k \setminus \{x\})$$

This translates to $H_k(V, V \setminus \{x\}) \cong H_n(\mathbb{R}^k, \mathbb{R}^k \setminus *)$ so that we now have

$$H_k(S^k, S^k \setminus \{x_i\}) \cong \mathbb{Z}$$

by proposition 6.2.2. However, there is an obvious isomorphism which makes the definition clearer.

Lemma 4.2.2 The local degree of a map $f : S^k \to S^k$ is well defined.

Proof By excision, we have an isomorphism

$$H_k(S^k, S^k \setminus \{x_i\}) \cong H_k(U_i, U_i \setminus \{x_i\})$$

by excising the piece $S^k \setminus U$.

By using the long exact sequence for relative homology, we have that

$$\cdots \longrightarrow H_k(S^k \setminus \{x_i\}) \longrightarrow H_k(S^k) \longrightarrow H_k(S^k, S^k \setminus \{x_i\}) \longrightarrow H_{k-1}(S^k \setminus \{x_i\}) \longrightarrow \cdots$$

But the first and latter terms are 0 since $S^k \setminus \{x_i\} \simeq \mathbb{R}^k$ so that we have an isomorphism

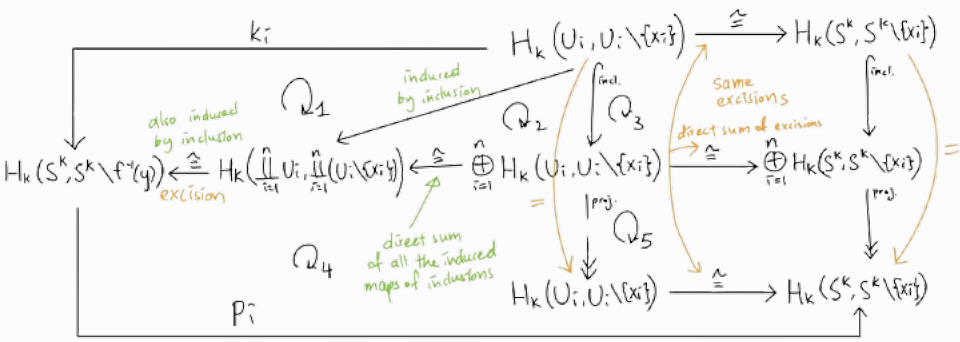
$$H_k(S^k) \cong H_k(S^k, S^k \setminus \{x_i\})$$

This is true for both the domain and the codomain of $f|_{x_i}$. ■

Proposition 4.2.3 Let $f : S^k \to S^k$ be a map. Let $f^{-1}(y) = \{x_1, \dots, x_n\}$. Then

$$\deg(f) = \sum_{i=1}^{n} \deg(f|_{x_i})$$

Proof Let U_i be an open neighbourhood of x_i for which $U_i \cap U_j = \emptyset$ for $i \neq j$. Let $f(U_i) \subseteq V$ for all i so that V is a neighbourhood of y. Consider the following diagram.

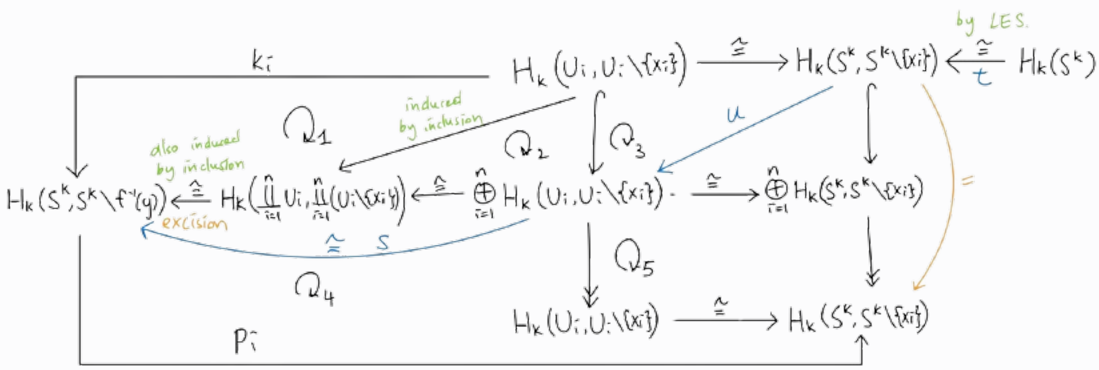


$k_i : H_k(U_i, U_i \setminus \{x_i\}) \to H_k(S^k, S^k \setminus f^{-1}(y))$ is by definition of the composition of the obvious maps and so the upper left part of the diagram is commutativity. The upper middle part of the diagram commutes since both are induce by inclusions an direct sum of the induced maps of inclusions. The upper right and lower right part are commutative since they are induced by the same excisions and the direct sum of the very same excisions. Finally, $p_i : H_k(S^k, S^k \setminus f^{-1}(y)) \to H_k(S^k, S^k \setminus \{x_i\})$ is defined to be the composition of the obvious maps and so the bottom left diagram is commutative by definition.

All of the above shows that the upper left of the following diagram commutes:

$$\begin{tikzcd}
 & H_k(U_i, U_i \setminus \{x_i\}) \arrow[r, "f_* = \times \deg(f|_{x_i})"] \arrow[dl, "\cong"'] \arrow[d, "k_i"] & H_k(V, V \setminus \{y\}) \arrow[d, "\cong"] \\
H_k(S^k, S^k \setminus \{x_i\}) & H_k(S^k, S^k \setminus f^{-1}(y)) \arrow[l, "p_i"'] \arrow[r, "f_*"] & H_k(S^k, S^k \setminus \{y\}) \\
 & H_k(S^k) \arrow[r, "f_* = \times \deg(f)"'] \arrow[u, "j_i"] \arrow[ul, "\cong"] & H_k(S^k) \arrow[u, "\cong"']
\end{tikzcd}$$

The bottom left of this diagram is commutative because:



and $s \circ u \circ t$ is the map j. The isomorphism on the top right is obtained by excision. k_i is induced by the inclusion $(S^k, S^k \setminus \{x_i\}) \hookrightarrow (S^k, S^k \setminus f^{-1}(y))$ and then by the very same excision

$$H_k(U_i, U_i \setminus \{x_i\}) \cong H_k(S^k, S^k \setminus \{x_i\})$$

so that the top right square commutes. The isomorphism on the bottom right is given by the long exact sequence for relative homology. It is commutative by the naturality of long exact sequences for relative homology.

By treating the middle term as

$$H_k(S^k, S^k \setminus f^{-1}(y))$$

one can see that k_i is just the inclusion into the direct summands. Similarly, using the first diagram of the proof we see that some part of p_i is a projection of the direct summands. Commutativity of the third diagram shows that $p_i(j(1)) = 1$. Since p_i is the projection from the direct summand, we must have that $p_i(0, \dots, 0, 1, 0, \dots, 0) = 1$ which shows that $j(1) = (1, \dots, 1)$. At the same time, k_i is the inclusion into the direct summands thus $(1, \dots, 1) = \sum_{i=1}^n k_i(1)$. Thus we conclude that $j(1) = \sum_{i=1}^n k_i(1)$.

Now recall that $\deg(f) = f_*(1)$. By the lower part of the second diagram, we have that

$$f_*(1) = f_*(j(1)) = f_*\left(\sum_{i=1}^n k_i(1)\right) = \sum_{i=1}^n f_*(k_i(1))$$

By the top part of the second diagram, we conclude that $f_*(k_i(1)) = \deg(f|_{x_i})$ for each i. Thus we have shown that

$$\deg(f) = \sum_{i=1}^n \deg(f|_{x_i})$$

and so we are done. ■

4.3 Cellular Homology

Singular homology is not often easy to calculate. A more combinatorial result comes from cellular homology, which actually gives the same result as singular homology.

Recall that CW complexes and subcomplexes always forms good pairs.

Lemma 4.3.1 Let X be a CW-complex with n-cells $\{D_\alpha^n\}$ for $n \geq 0$. Then

$$H_k(X^n, X^{n-1}) \cong \begin{cases} \oplus_\alpha \mathbb{Z} \cdot \{D_\alpha^n\} & \text{if } k = n \\ 0 & \text{otherwise} \end{cases}$$

Proof We have that

$$H_k(X^n, X^{n-1}) \cong \widetilde{H}_k(X^n / X^{n-1}) \cong \widetilde{H}_k\left(\bigvee_\alpha D_\alpha^n / \partial D_\alpha^n\right) \cong \bigoplus_\alpha \widetilde{H}_k(D_\alpha^n / \partial D_\alpha^n)$$

and so we conclude. ■

In fact, one can explicitly describe the isomorphism as follows. For D_α^n in the n-skeleton in X, it is isomorphic to Δ^n an n-simplex. Thus we have a continuous map

$$\Delta^n \cong D_\alpha^n \xrightarrow{\Phi_\alpha} X$$

which is in fact a relative cycle for the pair (X^n, X^{n-1}). Its relative homology class then generates the copy of $\mathbb{Z}$ corresponding to D_α^n.

Lemma 4.3.2 Let X be a CW-complex. Then the following are true regarding the singular homology of X.

- If X is of dimension k then $H_n(X) = 0$ for all $n > k$.
- The map $H_n(X^m) \to H_n(X)$ induced by the inclusion $X^m \to X$ is an isomorphism if $n < m$ and surjective if $m = n$.

Proof We proceed by induction. When $k = 0$, it is obvious. Assume the statement is true for $k - 1$. Consider the long exact sequence for relative homology:

$$\cdots \longrightarrow H_{n+1}(X^k, X^{k-1}) \longrightarrow H_n(X^{k-1}) \longrightarrow H_n(X^k) \longrightarrow H_n(X^k, X^{k-1}) \longrightarrow \cdots$$

By induction, the second term vanishes for $n > k$. By the previous lemma, the last term vanishes for $n > k$. Thus we obtain the required isomorphisms.

Similarly, by the above long exact sequence, when $n < k - 1$, both outer and terms vanish and the middle arrow becomes an isomorphism. For $n < k$, the last term vanishes and the middle arrow is a surjection. Thus for $n < m$, we have isomorphisms:

$$H_n(X^m) \longrightarrow H_n(X^{m+1}) \longrightarrow H_n(X^{m+2}) \longrightarrow \cdots$$

By properties of the geometric realization, we must have that every compact subspace of X must lie inside some X^t for $t \in \mathbb{N}$. For any $\tau \in C_{n+1}(X)$, we thus have $\tau \in C_{n+1}(X^t)$ for some $t > n$. Hence τ maps to 0 in $H_n(X^t)$ if τ is an n-cycle. Thus all these groups must be equal to $H_n(X)$. On the other hand, if $n = m$, then the first arrow $H_n(X^m) \to H_n(X^{m+1})$ is only surjective while the remaining ones are isomorphisms. Thus we are done. ■

Using the above lemma, we now have a spliced diagram :

$$\begin{tikzcd}
 & & & 0 & \\
0 \arrow[dr] & & H_n(X^{n+1}) \arrow[ur] & & \\
 & H_n(X^n) \arrow[ur] \arrow[dr, "\beta"] & & & \\
H_{n+1}(X^{n+1}, X^n) \arrow[ur, "\alpha"] \arrow[rr, "d_{n+1}"] & & H_n(X^n, X^{n-1}) \arrow[rr, "d_n"] \arrow[dr, "\gamma"] & & H_{n-1}(X^{n-1}, X^{n-2}) \\
 & & & H_{n-1}(X^{n-1}) \arrow[ur, "\delta"] & \\
 & & 0 \arrow[ur] & &
\end{tikzcd}$$

where the diagonals come from the long exact sequence in the proof of the first part of lemma 7.4.3. The d_n is then defined through the composition of the diagonal arrows.

Definition 4.3.3 (Cellular Boundary)
Let X be a CW complex. Let $\alpha : H_{n+1}(X^{n+1}, X^n) \to H_n(X^n)$ be the induced map from the long exact sequence of relative homology. Let $\beta : H_n(X^n) \to H_n(X^n, X^{n-1})$ be the quotient map in the long exact sequence of relative homology. Define the cellular boundary map of X to be $d_{\text{CW}} = \beta \circ \alpha$.

Lemma 4.3.4
Let X be a CW complex. Then we have $d_{\text{CW}} \circ d_{\text{CW}} = 0$.

Proof
The composition $d_n \circ d_{n+1}$ factors through the two blue arrows in the diagram. They are part of a long exact sequence and so their composition is 0. ■

Recall that a CW-complex X is defined recursively through the use of attaching maps $\{\phi_\alpha : S_\alpha^{n-1} \to$

$X^{n-1}|\alpha \in I_n\}$ for each n, such that

$$X^n = \left(X^{n-1} \amalg \coprod_{\alpha \in I_n} D^n_\alpha \right) / \sim$$

where the equivalence relation is $x \sim \phi_\alpha(x)$ for all $x \in \partial D^n_\alpha$ (This is an amalgamated product over $\phi_\alpha(x)$). Notice that we have

$$\frac{X^n}{X^{n-1}} \cong \bigvee_{\alpha \in I_n} \frac{D^n_\alpha}{\partial D^n_\alpha} = \bigvee_{\alpha \in I_n} S^n_\alpha$$

implies that there are canonical quotient maps obtain by collapsing all sphere other than one into a single point by an equivalence relation:

$$\pi_\alpha : X^n \twoheadrightarrow \tfrac{X^n}{X^{n-1}} \cong \textstyle\bigvee_{\alpha \in I_n} S^n_\alpha \twoheadrightarrow S^n_\alpha$$

Theorem 4.3.5 (Cellular Boundary Formula) Let X be a CW-complex with attaching maps denoted ϕ_α, and $C^{\text{CW}}_\bullet(X)$ the cellular chain complex of X with generators $\{[D^n_\alpha]|_{\alpha \in I_n}\}$ for each degree n. Then the cellular boundary is given as follows.

- In degree $n = 1$, we have
$$d_1\left([D^1_\alpha]\right) = [\phi_\alpha(1)] - [\phi_\alpha(0)]$$
- In degrees $n > 1$, we have
$$d_n\left([D^n_\alpha]\right) = \sum_{\beta \in I_{n-1}} d_{\alpha\beta} \cdot [D^{n-1}_\beta]$$
where
$$d_{\alpha\beta} = \deg\left(\Delta_{\alpha\beta} : S^{n-1}_\alpha \overset{\phi_\alpha}{\to} X^{n-1} \overset{\pi_\beta}{\to} S^{n-1}_\beta\right)$$

On computation, one crucial fact to notice is that the map π_β is a projection to the quotient topology. This means that after applying π_β, the map

$$\Delta_{\alpha\beta} = \pi_\beta \circ \phi_\alpha : S^{n-1}_\alpha \to S^{n-1}_\beta$$

forgets about every boundary of the attached discs to X^n other than the boundary of the disc D^n_β. Also notice that $\alpha \in I_n$ loops over all attaching maps and disks on the nth dimension, while $\beta \in I_{n-1}$ loops over that of the $n-1$ dimension. Notice that in the formula, β is used to indicate the S^{n-1} which are of dimension n. This makes sense because in π_β, there is a projection $X^{n-1} \to X^{n-1}/X^{n-2}$.

$$\frac{X^{n-1}}{X^{n-2}} = \bigvee_{\beta \in I_{n-1}} \frac{D^{n-1}_\beta}{\partial D^{n-1}_\beta}$$

is then a wedge sum of all disks in dimension $n-1$ modulo boundary.

Definition 4.3.6 (Cellular Chain Complex)
Let X be a CW complex. Define the cellular chain complex $C^{\text{CW}}_\bullet(X)$ as follows.

- For each n, define $C^{\text{CW}}_n = H_n(X^n, X^{n-1}) \cong \bigoplus_\alpha \mathbb{Z} \cdot \{D^n_\alpha\}$ where D^n_α is an n-cell of X.
- Define the differential d to be the cellular boundary.

Definition 4.3.7
Let X be a CW complex. Define the nth cellular homology of X to be

$$H^{\text{CW}}_n(X) = H_n(C^{\text{CW}}_\bullet(X), d_{\text{CW}})$$

Lemma 4.3.8
For any CW-complex X, there are canonical isomorphisms

$$H^{\text{CW}}_n(X) \cong H_n(X)$$

Proof Since δ is injective, we have that

$$\ker(d_n) = \ker(\gamma) = \operatorname{im}(\beta)$$

Also by lemma 7.3.2, we have that $H_n(X^{n+1}) \cong H_n(X)$. This means that

$$H_n(X) \cong \text{coker}(\alpha) \cong \frac{\operatorname{im}(\beta)}{\operatorname{im}(\beta \circ \alpha)} = \frac{\ker(d_n)}{\operatorname{im}(d_{n+1})}$$

where the second isomorphism comers from the fact that β is injective. ∎

We can now compute the singular homology of more complicated spaces such as the projective space.

Example 4.3.9 Let $n \geq 1$. Then the real projective space $\mathbb{RP}^n$ has the following homology groups:

$$H_k(\mathbb{RP}^n) = \begin{cases} \mathbb{Z} & \text{if } k = 0 \text{ and } k = n \text{ is odd} \\ \mathbb{Z}/2\mathbb{Z} & \text{if } 0 < k < n \text{ and } k \text{ is odd} \\ 0 & \text{otherwise} \end{cases}$$

Proof Recall that the CW complex structure of $\mathbb{RP}^n$ consists of one k-cell for each $0 \leq k \leq n$. We have to determine the boundary operator. This is done by induction.

The base case is obtained easily since $S^1 \cong \mathbb{RP}^1$.

Suppose that the attaching maps are defined for all lower values of k. Recall that the attaching map of each k-cell is defined by the two-to-one map $f : S^{k-1} \to \mathbb{RP}^{k-1}$. This forms the first part of the map $\Delta_{\alpha\beta}$ of the formula for the cellular boundary maps. To proceed, we quotient out the X^{k-2} skeleton to obtain

$$S^{k-1} \to \mathbb{RP}^{k-1} \to \frac{\mathbb{RP}^{k-1}}{\mathbb{RP}^{k-2}} \cong S^{k-1}$$

We compute the degree of this map using local degrees. Since it is a two-to-one covering map, we concern ourselves with the north / south pole of S^{k-1}. Choose your favourite homeomorphism $D^{k-1}/\partial D^{k-1} \cong S^{k-1}$. Then the map $f : S^{k-1} \to \mathbb{RP}^{k-1}$ is a local homeomorphism since it is a covering space. Thus the degree of the map is ± 1, fixed by the choice of the homeomorphism. This similar for the south pole. In fact, the map is identical to that of the local homeomorphism at the north pole and so the local homeomorphism is just the antipodal map composed with the local homeomorphism at the north pole.

Denote N and S the north and south pole respectively. Since $\deg(f|_N) = \pm 1$ is fixed and the antipodal map has degree $(-1)^k$, we obtain

$$\deg(f) = \deg(f|_N) + (-1)^n \deg(f|_N) = \begin{cases} \pm 2 & \text{if } n \text{ is even} \\ 0 & \text{if } n \text{ is odd} \end{cases}$$

Thus the cellular chain complex together with induction, is of the form

$$0 \longrightarrow \mathbb{Z} \xrightarrow{d_k} \mathbb{Z} \xrightarrow{d_{k-1}} \cdots \xrightarrow{\pm 2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{\pm 2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \longrightarrow 0$$

where $d_k = \pm 2$ if k is even and $d_k = 0$ if k is odd. It follows that

$$\deg(f) = \deg(f|_N) + (-1)^n \deg(f|_N) = \begin{cases} \pm 2 & \text{if } n \text{ is even} \\ 0 & \text{if } n \text{ is odd} \end{cases}$$

∎

Example 4.3.10
Let X be the space obtained by attaching an $(n+1)$-cell to S^n by a map $f: S^n \to S^n$ of degree p. Then we have

$$H_k(X) = \begin{cases} \mathbb{Z} & \text{if } k = 0 \\ \mathbb{Z}/p\mathbb{Z} & \text{if } n = k \\ 0 & \text{otherwise} \end{cases}$$

Proof
Using cellular homology, we compute that $d_{\text{CW}}([D^{n+1}]) = \deg(S^n \to X^n = S^n) \cdot [D^n] = p \cdot [D^n]$. So we have $H_n(X) = \frac{\ker(\mathbb{Z}[D^n] \to 0)}{\operatorname{im}(\mathbb{Z}[D^{[}n+1] \to \mathbb{Z}[D^n])} = \mathbb{Z}/p\mathbb{Z}$. Since the map $\mathbb{Z}[D^{n+1}] \to \mathbb{Z}[D^n]$ is injective, we have $H_{n+1}(X) = 0$. ■

Example 4.3.11
Let X be the space obtained by attaching two $(n+1)$-cells to S^n by a maps $f: S^n \to S^n$ of degree p and q respectively. Then we have

$$H_k(X) = \begin{cases} \mathbb{Z} & \text{if } k = 0, n+1 \\ \mathbb{Z}/\gcd(p,q)\mathbb{Z} & \text{if } k = n \\ 0 & \text{otherwise} \end{cases}$$

Proof
Using a similar strategy, we note that $C_{n+1}^{\text{CW}}(X) \to C_n^{\text{CW}}(X)$ is the map $\mathbb{Z}^2 \to \mathbb{Z}$ given by $(1,0) \mapsto p$ and $(0,1) \mapsto q$. Then we have $H_n(X) = \frac{\mathbb{Z}}{\gcd(p,q)\mathbb{Z}}$ and the $H_{n+1}(X) = \ker(\mathbb{Z}^2 \overset{(p,q)}{\to} \mathbb{Z})$ is infinite cyclic. ■

5 Variants of Homology

5.1 Homology with Coefficients

We would now like to generalize singular homology so that the coefficients of the chain complex does not take values in $\mathbb{Z}$, but instead in an arbitrary group G. For this, we first modify the group of n-chains.

Definition 5.1.1 (Singular n-Chains with Coefficients in a Group) Let X be a space. Let G be an abelian group. Define the group of singular n-chains with coefficients in G to be

$$C_n(X;G) = C_n(X) \otimes G$$

Let $A \subseteq X$ be a subspace. Define the relative n-chains with coefficients in G to be the group

$$C_n(X,A;G) = \frac{C_n(X;G)}{C_n(A;G)}$$

In particular, notice that the tensor product here just means that $C_n(X;G)$ is just copies of G, where the number of copies is precisely the number of singular n-simplexes in X.

Definition 5.1.2 (Boundary Operator) Let X be a space. Let G be an abelian group. Define the boundary operator of the singular chain complex with coefficients in G to be

$$\partial_G = \partial_n \otimes 1_G : C_n(X;G) \to C_{n-1}(X;G)$$

defined by $\sigma \otimes g \mapsto \partial_n(\sigma) \otimes g$ and then extended linearly.

Lemma 5.1.3 Let X be a space. Let G be an abelian group. Then $\partial_G \circ \partial_G = 0$. Moreover, the groups $C_n(X;G)$ together with the boundary operator ∂_G is a chain complex.

Definition 5.1.4 (Singular Homology with Coefficients) Let X be a space. Let G be an abelian group. Define the singular homology groups of X to be the homology groups of the chain complex $(C_\bullet(X;G), \partial_G)$. This means that

$$H_n(X;G) = \frac{\ker(\partial_G : C_n(X;G) \to C_{n-1}(X;G))}{\operatorname{im}(\partial_G : C_{n+1}(X;G) \to C_n(X;G))} = H_n(C_\bullet(X;G))$$

for $n \in \mathbb{N}$.

A natural question would be to ask whether there is a relation between homology with integer coefficients $H_n(X)$ and homology with arbitrary coefficients $H_n(X;G)$. Namely, the chain complex for singular homology with coefficients simply replaces the integral coefficients of the original singular chain complex via the tensor product. Is there a natural association between $H_n(X) \otimes G$ and $H_n(X;G)$?

The answer is yes, but it is not an isomorphism.

Theorem 5.1.5 (Universal Coefficient Theorem) Let (X,A) be a pair of space. Let T be an abelian group. Then there exists a natural map $h : H_n(X,A) \otimes T \to H_n(X,A;T)$ such that $\operatorname{coker}(h) \cong \operatorname{Tor}_1^{\mathbb{Z}}(H_{n-1}(X,A),T)$ and a split exact sequence (that is not natural) of the form

$$0 \longrightarrow H_n(X,A) \otimes T \xrightarrow{h} H_n(X,A;T) \longrightarrow \operatorname{Tor}_1^{\mathbb{Z}}(H_{n-1}(X,A),T) \longrightarrow 0$$

for any $n \in \mathbb{N}$. In particular, split exactness implies that there is an isomorphism

$$H_n(X,A;T) \cong H_n(X,A) \otimes T \oplus \operatorname{Tor}_1^{\mathbb{Z}}(H_{n-1}(X,A),T)$$

for any $n \in \mathbb{N}$.

5.2 Kunneth Theorem for Homology

Theorem 5.2.1 (The Eilenberg-Zilbur Theorem)
Let X, Y be spaces. Then $C_\bullet(X) \otimes_\mathbb{Z} C_\bullet(Y)$ and $C_\bullet(X \times Y)$ are chain homotopy equivalent, that is natural in X and Y.

Proposition 5.2.2 Let X and Y be CW-complexes. Let R be a principal ideal domain. Then there is a short exact sequence

$$0 \longrightarrow \bigoplus_{i+j=n} H_i(X;R) \otimes_R H_j(Y;R) \xrightarrow{\times} H_n(X \times Y;R) \longrightarrow \bigoplus_{i+j=n} \operatorname{Tor}_1^R(H_i(X;R), H_{j-1}(Y;R)) \longrightarrow 0$$

induced by the cross product, that is natural in maps $f : X \to A$ and $g : Y \to B$. Moreover, this sequence splits.

5.3 Relation to Simplicial Homology

Theorem 5.3.1 Let X be a topological space endowed with a Δ-complex structure $(T, |T| \cong X)$. The induced map $H_n^\Delta(X) \to H_n(X)$ by the map $s : \Delta^n \to X$ for each $s \in T^n$ is an isomorphism.

Proof Firstly, every n-simplex $s \in T$ induces a canonical continuous map $s : \Delta^n \to X$. This extends to a homomorphism $\Delta_n(T) \to C_n(X)$ and so to a chain map and descends to homology.

Denote T^k the Δ-sets of simplices in T of dimension at most k. Define $\Delta_\bullet(T^k, T^{k-1}) = \Delta_\bullet(T^k)/\Delta_\bullet(T^{k-1})$ and accordingly, $H_n(\Delta_\bullet(T^k, T^{k-1}))$. Then the short exact sequence of chain complexes

$$0 \longrightarrow \Delta_\bullet(T^{k-1}) \longrightarrow \Delta_\bullet(T^k) \longrightarrow \Delta_\bullet(T^k, T^{k-1}) \longrightarrow 0$$

Together with the inclusion maps and by naturality in theorem 1.4.3, give the following:

$$\begin{tikzcd}
\cdots \arrow[r] & H_{n+1}(T^k, T^{k-1}) \arrow[r] \arrow[d] & H_n(T^{k-1}) \arrow[r] \arrow[d] & H_n(T^{k-1}) \arrow[r] \arrow[d] & H_n(T^k, T^{k-1}) \arrow[r] \arrow[d] & H_{n-1}(T^{k-1}) \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & H_{n+1}(|T^k|, |T^{k-1}|) \arrow[r] & H_n(|T^{k-1}|) \arrow[r] & H_n(|T^{k-1}|) \arrow[r] & H_n(|T^k|, |T^{k-1}|) \arrow[r] & H_{n-1}(|T^{k-1}|) \arrow[r] & \cdots
\end{tikzcd}$$

When $k = 0$, $|T^0|$ is a discrete topological space on the set T^0, and the map $H_n(T^0) \to H_n(|T^0|)$ is an isomorphism. By induction and the five lemma, the middle map is an isomorphism provided that the first and fourth maps are isomorphisms.

Note that $\Delta_\bullet(T^{k-1})$ is a chain complex in degrees $k-1, k-2$, down to 0. And $\Delta_\bullet(T^k)$ is the same chain complex with an additional term $\mathbb{Z}T_k$ in degree k. It follows that $\Delta_\bullet(T^k, T^{k-1})$ is the chain complex with $\mathbb{Z}T^k$ concentrated in degree k. In particular, we have that

$$H_n(T^k, T^{k-1}) = \begin{cases} \mathbb{Z}T^k & \text{if } n = k \\ 0 & \text{otherwise} \end{cases}$$

By construction of the geometric realization, we have a homeomorphism

$$\frac{|T^k|}{|T^{k-1}|} \cong \frac{\Delta^k \times T^k}{\partial \Delta^k \times T^k} \cong \bigvee_{T_k} \frac{\Delta^k}{\partial \Delta^k}$$

Using lemma 6.2.4 and lemma 6.2.5, we have that

$$H_n(|T^k|, |T^{k-1}|) \cong \widetilde{H}_n(|T^k|/|T^{k-1}|) \cong \begin{cases} \mathbb{Z}T_k & \text{if } n = k \\ 0 & \text{otherwisev} \end{cases}$$

with generators corresponding to elements $s \in T^k$ via the relative cycles $s : \Delta^k \to |T^k|$. It follows that an isomorphism occurs in the first and fourth position of the long exact sequence of relative homology groups.

If $T = T^n$ for some $n \in \mathbb{N}$, we are done.

Continuing the proof, let $z \in Z_n(T)$ be an n-cycle whose image in $H_n(|T|)$ vanishes. Then there exists $\tau \in C_{n+1}(|T|)$ with $\partial\tau = z$. Now since $|T|$ is the geometric realization, we must have every compact subspace of $|T|$ lying in some $|T^k|$. Hence $\tau \in C_{n+1}(|T^k|)$ for some $k > n$. We deduce that z maps to 0 in $H_n(|T^k|)$. By the previous argument, we have taht $z = 0 \in H_n(T^k) = H_n(T)$. This shows injectivity of $H_n(T) \to H_n(|T|)$.

Similarly, let $\sigma \in Z_n(T)$ be an n-cycle. We have seen that $\sigma \in Z_n(|T^k|)$ for some $k > n$ and by the argument above, $[\sigma]$ comes from $H_n(T^k) = H_n(T)$. This shows surjectivity of $H_n(T) \to H_n(|T|)$ and so we are done. ■

We can finally show that simplicial homology is independent of the choice of Δ-complex structure.

Corollary 5.3.2 The simplicial homology $H_\bullet^\Delta(X)$ depends on X only and not on the Δ-complex structure.

Proof Indeed the geometric realization of any two Δ-complex structure is isomorphic to the same singular homology group. ■

Corollary 5.3.3 Suppose X has a Δ-complex structure with simplicies in dimension $\leq k$ only. Then $H_n(X) = 0$ for all $n > k$.

Proof Direct since singular homology coincides with simplicial homology. ■

5.4 Axioms for Unreduced Homology

Definition 5.4.1 (Generalized Homology Theory for CW Pairs) A Generalized Homology Theory is a collection of functors and natural transformations

$$h_n : \mathbf{CW}^2 \to \mathbf{Ab} \quad \text{and} \quad \delta_n : h_n \Rightarrow h_n \circ F$$

where $F(X, Y) = (Y, \emptyset)$, for each $n \in \mathbb{N}$, such that the following are true.

- Homotopy Invariance: If $f \simeq g : (X, A) \to (Y, B)$ then

$$h_n(f) = h_n(g) : h_n(X, A) \to h_n(Y, B)$$

- Exactness: There exists an exact sequence

$$\cdots \longrightarrow h_{n+1}(X, A) \xrightarrow{\delta_{n+1}} h_n(A, \emptyset) \xrightarrow{h_n(i)} h_n(X, \emptyset) \xrightarrow{h_n(j)} h_n(X, A) \xrightarrow{\delta_n} h_{n-1}(A, \emptyset) \longrightarrow \cdots$$

  where $i : (A, \emptyset) \to (X, \emptyset)$ and $j : (X, \emptyset) \to (X, A)$ are inclusions.
- Excision: If $\overline{E} \subseteq A^\circ \subseteq X$, then there is an isomorphism

$$h_n(X \setminus E, A \setminus E) \cong h_n(X, A)$$

  induced by the inclusion map.

The homotopy invariance axiom means that homology descends to a functor $\mathbf{HoCW}^2 \to \mathbf{Ab}$. Therefore some authors write the axiom implicit in the definition of the functors h_n, and then saying that a homology theory consists of functors $h_n : \mathbf{HoCW} \to \mathbf{Ab}$ instead.

Lemma 5.4.2 The excision axiom is equivalent to saying that $X = A^\circ \cup B^\circ$ with inclusion map $\iota : (B, A \cap B) \to (X, A)$ implies $h_n(\iota) : h_n(B, A \cap B) \to h_n(X, A)$ is an isomorphism.

Definition 5.4.3 (Additive Homology Theory) Let $h_n : \mathbf{CW}^2 \to \mathbf{Ab}$ be a generalized homology theory. We say that h_n is additive if the following axiom is satisfied.

- Additivity: If $(X, A) = \coprod_{i\in I}(X_i, A_i)$, then the direct sum of the inclusion maps

$$\bigoplus_{i\in I} h_n(X_i, A_i) \cong h_n(X, A)$$

  is an isomorphism

We mention for once and for all that the additivity axiom is required only when the CW complexes are non-finite. In particular, in order for the homology theory to be meaningful, we must restrict the underlying category of spaces to be finite CW complexes if one drops the additivity axiom.

6 Singular Cohomology

6.1 Singular Cohomology

We now have all the pieces to define singular cohomology. This is a three step functorial process so that the end result is also functorial.

Definition 6.1.1 (The Singular Cochain Complex) Let X be a space. Let G be an abelian group. Define the singular cochain complex $(C^\bullet(X;G),\partial^\bullet)$ with coefficients in G to be the cochain complex given by the following data.

- The nth term is given by
$$C^n(X;G)=\text{Hom}_\mathbb{Z}(C_n(X);G)$$
the group of singular n-cochains of X.
- The boundary operator $\partial^n:C^{n-1}(X;G)\to C^n(X;G)$ is given by the formula
$$\partial^n(\phi)=\phi\circ\partial_n$$
for $\phi\in C^{n-1}(X;G)$.

It is the dual cochain complex of the singular chain complex $(C_\bullet(X),\partial_\bullet)$ of X.

Definition 6.1.2 (Singular Cohomology Groups of a Space) Let X be a topological space. Let G be a group. The nth singular cohomology group of X with coefficients in G is defined to be
$$H^n(X;G)=H_n(C^\bullet(X;G),\partial^\bullet)=\frac{\ker(\partial^{n+1}:C^n(X;G)\to C^{n+1}(X;G))}{\text{im}(\partial^n:C^{n-1}(X;G)\to C^n(X;G))}$$

Recall that for a space X, its singular chain complex $C_\bullet^{\text{Sing}}(X)$ is defined to be free abelian group
$$C_n^{\text{Sing}}(X)=\mathbb{Z}[\Delta^n]$$
on the set of singular n-simplicies on X. By dualizing it, one obtains the singular cochain complex
$$C^n_{\text{Sing}}(X)=\text{Hom}(\mathbb{Z}[\Delta^n],G)$$
for any abelian group G. Such a map $f\in\text{Hom}(\mathbb{Z}[\Delta^n],G)$ is determined on the generators of the free abelian group by its universal property. And so f is essentially determined by the choice of an element $g\in G$ for every singular n-simplex. In other words, every element of $C^n_{\text{Sing}}(X;G)$ of X is just a labelling of every singular n-simplex with an element of G. Now if $f:\mathbb{Z}[\Delta^n]\to G$ is a cocycle, we simply require that the sum of the values that f takes on the boundary of Δ^n to be 0. In particular,

- Elements of $C^0_{\text{Sing}}(X;G)$ simply labels every point of X with an element of G.
- Elements of $Z^0(X;G)$ labels every point of X with an element of G such that $f(v_0)=f(v_1)$, thus forming a 1-simplex $v:[0,1]\to X$ (Indeed, if $f\in Z^0(X;G)$, then $\delta(f)=f\circ\partial=0$. Choosing the 1-simplex $[0,1]$ shows that $0=f(\partial([0,1]))=f(v_0-v_1)=f(v_0)-f(v_1)$ which is exactly what we wanted). But this means that elements of $Z^0(X;G)$ labels every point of X with an element of G such that f is constant on path components of X.

Definition 6.1.3 (Induced Homomorphisms) Let $f:X\to Y$ be a continuous map. Let G be an abelian group. Define the induced map on cohomology
$$f^*:H^n(Y;G)\to H^n(X;G)$$
by the formula $[\alpha]\mapsto[\alpha\circ f]$.

For an R-module M, we can describe the construction of singular cohomology groups as a three step functorial process
$$H^n(-;M):\textbf{Top}\xrightarrow{C_\bullet(-)}\textbf{Ch}(R)\xrightarrow{\text{Hom}_R(-,M)}\textbf{CCh}(\mathbb{Z})\xrightarrow{H_n(-)}\textbf{Ab}$$

Therefore the construction of singular cohomology groups are also functorial. In the case that R is a commutative ring, we may upgrade the three step process as

$$H^n(-;M) : \textbf{Top} \overset{C_\bullet(-)}{\longrightarrow} \textbf{Ch}(R) \overset{\text{Hom}_R(-,M)}{\longrightarrow} \textbf{CCh}(R) \overset{H_n(-)}{\longrightarrow} {}_R\textbf{Mod}$$

Theorem 6.1.4 (Functoriality of Cohomology) Let X, Y be spaces. Let $f : X \to Y$ be a continuous map. Let G be an abelian group. Then the induced homomorphism

$$f^* : H^n(Y;G) \to H^n(X;G)$$

is well defined and is a group homomorphism. Moreover, the following are true.

- If Z is a space and $g : Y \to Z$ is another map, then

$$(g \circ f)^* = f^* \circ g^*$$

- We have that

$$(\text{id}_X)^* = \text{id}_{H^n(X;G)}$$

Proof As seen above cohomology is a three step functorial process, hence the induced map satisfies all the functorial properties. ∎

Theorem 6.1.5 (Homotopy Invariance) Let X, Y be spaces. Let $f, g : X \to Y$ be continuous maps. Let G be an abelian group. If f and g are homotopic, then they induce the same map

$$f^* = g^* : H^n(Y;G) \to H^n(X;G)$$

on singular cohomology.

Proof We have seen that in the proof of 4.4.3, $f_*, g_* : C_n(X) \to C_n(Y)$ are chain homotopic by some chain homotopy $\eta : C_n(X) \to C_{n+1}(Y)$. This means that have

$$g_* - f_* = \partial \circ \eta + \eta \circ \partial : C_n(X) \to C_n(Y)$$

Both maps are group homomorphisms hence we can dualize and applying the hom functor to get

$$g^* - f^* = \partial^* \circ \eta^* + \eta^* \circ \partial^*$$

This means that the maps$f^*, g^* : C^n(X;G) \to C^n(Y;G)$ are chain homotopic, hence they induce the same map on cohomology. ∎

Theorem 6.1.6 (Mayer-Vietoris Sequence) Let X be a topological space and U_1, U_2 be open sets of X such that $X = U_1 \cup U_2$ and that $U_1 \cap U_2 \neq \emptyset$. Write $i_1 : U_1 \cap U_2 \to U_1$, $i_2 : U_1 \cap U_2 \to U_2$, $j_1 : U_1 \to X$ and $j_2 : U_2 \to X$ the inclusion maps. Let G be an abelian group. Then there is a long exact sequence

$$\cdots \longrightarrow H^{n-1}(X;G) \overset{\partial}{\longrightarrow} H^n(U_1 \cap U_2;G) \overset{(i_1)^* - (i_2)^*}{\longrightarrow} H^n(U_1;G) \oplus H^n(U_2;G) \overset{(j_1)^* + (j_2)^*}{\longrightarrow} H^n(X;G) \overset{\partial}{\longrightarrow} H^{n+1}(U_1 \cap U_2;G) \longrightarrow \cdots$$

in cohomology.

Proof ∎

Definition 6.1.7 (Reduced Cohomology Groups) Let X be a space. Let G be an abelian group. Define the reduced cohomology groups

$$\widetilde{H}^n(X;G)$$

of X with coefficients in G to be the nth cohomology groups of the augmented singular chain complex.

Proposition 6.1.8 Let X be a space. Let R be a ring. Let M be an R-module. Then the following are true regarding the reduced cohomology of X.

- For all $n > 0$,
$$\widetilde{H}^n(X; M) = H^n(X; M)$$
- When $n = 0$, the universal coefficient theorem induces an isomorphism
$$\widetilde{H}^0(X; M) \cong \operatorname{Hom}(\widetilde{H}_0(X), M)$$

Example 6.1.9 Let $n \in \mathbb{N}$. Let A be an abelian group. The singular cohomology of S^n with coefficients in A is given by
$$H^k(S^n; A) = \begin{cases} A & \text{if } k = 0 \text{ or } n \\ 0 & \text{otherwise} \end{cases}$$

Example 6.1.10 Let $n \geq 1$. Let A be an abelian group. The singular cohomology of $\mathbb{RP}^n$ with coefficients in A is given by
$$H^k(\mathbb{RP}^n; A) = \begin{cases} A & \text{if } k = 0 \\ A/2A & \text{if } 0 < k < n \text{ and } k \text{ even} \\ \{a \in A \mid 2 \text{ divides } |a|\} & \text{if } 0 < k < n \text{ and } k \text{ odd} \\ 0 & \text{otherwise} \end{cases}$$

Example 6.1.11 Let $n \geq 1$. Let A be an abelian group. The singular cohomology of $\mathbb{CP}^n$ with coefficients in A is given by
$$H^k(\mathbb{CP}^n; A) = \begin{cases} A & \text{if } 0 \leq k \leq 2n \text{ even} \\ 0 & \text{otherwise} \end{cases}$$

6.2 Universal Coefficient Theorem

Theorem 6.2.1 (Universal Coefficient Theorem)
Let R be a PID. Let M be an R-module. Let X be a space. Then there is a split exact sequence (that does not split naturally) of the form
$$0 \longrightarrow \operatorname{Ext}^1_R(H_{n-1}(X; R)), M \longrightarrow H^n(X; M) \xrightarrow{h} \operatorname{Hom}(H_n(X; R), M) \longrightarrow 0$$
where h is the map defined by $[\alpha] \mapsto \left[\overline{\alpha|_{Z_n(X)}}\right]$.

6.3 Relative Cohomology

Definition 6.3.1 (Relative Cochain Complex) Let X be a space and let $A \subseteq X$ be a subspace. Let G be an abelian group. Define the relative cochain complex $(C^\bullet(X, A; G), \partial^*)$ with coefficients in G to be given by the following data.

- The nth term is given by
$$C^n(X, A; G) = \operatorname{Hom}_{\mathbb{Z}}(C_n(X, A), G)$$
the group of singular n-cochains relative to A.
- The boundary operator $\partial^n : C^{n-1}(X; G) \to C^n(X; G)$ is given by the formula
$$\partial^n(\phi) = \phi \circ \partial_n$$
for $\phi \in C^{n-1}(X, A; G)$.

It is the dual cochain complex of the relative singular chain complex $(C_\bullet(X, A), \partial_\bullet)$ of X with respect to A.

Lemma 6.3.2 Let X be a space and let $A \subseteq X$ be a subspace. Let G be an abelian group. Then the nth term of the relative cochain complex is given by

$$C^n(X, A; G) = \{\alpha \in C^n(X; G) \mid \alpha(\sigma) = 0 \text{ for all } \operatorname{im}(\sigma) \subseteq A\}$$

Definition 6.3.3 (Relative Cohomology Groups of a Space) Let X be a topological space. Let G be an abelian group. The nth relative cohomology group of X with coefficients in G is defined to be

$$H^n(X, A; G) = H_n(C^\bullet(X, A; G), \partial^\bullet) = \frac{\ker(\partial^{n+1} : C^n(X, A; G) \to C^{n+1}(X, A; G))}{\operatorname{im}(\partial^n : C^{n-1}(X, A; G) \to C^n(X, A; G))}$$

TBA: Functoriality

Theorem 6.3.4 (Excision) Let (X, A) be a pair of spaces. Let $Z \subseteq A$ be a subspace such that $\overline{Z} \subseteq A^\circ$. Let G be an abelian group. Then the inclusion map $\iota : (X \setminus Z, A \setminus Z) \hookrightarrow (X, A)$ induces an isomorphism

$$\iota^* : H^n(X, A; G) \xrightarrow{\cong} H^n(X \setminus Z, A \setminus Z; G)$$

in cohomology groups for all $n \in \mathbb{N}$.

Theorem 6.3.5 (Long Exact Sequence of Relative Cohomology Groups) Let X be a space and let $A \subseteq X$ be a subspace. Let G be an abelian group. Then there is an exact sequence

$$\cdots \longrightarrow H^{n-1}(A; G) \xrightarrow{\partial} H^n(X, A; G) \xrightarrow{j^*} H^n(X; G) \xrightarrow{i^*} H^n(A; G) \longrightarrow \cdots$$

where $\iota : A \to X$ is the inclusion map, $j : X \to X \setminus A$ is the quotient map and ∂ is the connecting homomorphism induced by the snake lemma.

Proof Notice that

$$0 \longrightarrow C^\bullet(X, A; G) \xrightarrow{j^*} C^\bullet(X) \xrightarrow{i^*} C^\bullet(A) \longrightarrow 0$$

is a short exact sequence by the above lemma. Thus it induces a long exact sequence in cohomology groups.

Consider $[z] = z + C_n(A) \in C_n(X, A)$ a cycle in $C_n(X, A)$. The surjective map j is the projection map so $j(z) = [z]$. Now $d(z) \in \ker(j)$ because

$$j(d(z)) = d(j(z)) = d([z]) = 0$$

by assumption So $d(z) \in \ker(j) = \operatorname{im}(i)$. Since i is the inclusion, $[d(z)] \in C_{n-1}(A)$ is precisely the element that ∂ maps $[z]$ to. ■

Theorem 6.3.6 (Long Exact Sequence of Triplet of Spaces) Let X be a space. Let $B \subseteq A \subseteq X$ be subspaces. Let $j : B \hookrightarrow A$ and $i : A \hookrightarrow X$ be inclusions. Let M be a R-module for some ring R. Then there is a long exact sequence in cohomology

$$\cdots \longrightarrow H^{n-1}(A, B; M) \xrightarrow{\delta} H^n(X, A; M) \xrightarrow{j^*} H^n(X, B; M) \xrightarrow{i^*} H^n(A, B; M) \xrightarrow{\delta} H^{n+1}(X, A; M) \longrightarrow \cdots$$

induced by the dual of the short exact sequence of the above proposition. The map δ is given by the connecting homomorphism.

The connecting homomorphism ∂ in homology is related to the connecting homomorphism δ in the following way.

Proposition 6.3.7 Let (X, A) be a pair of space. Let M be a R-module for some ring R. Let $\partial : H_{n+1}(X, A) \to H_n(A)$ denote the connecting homomorphism in relative homology. Let $\delta : H^n(A; M) \to H^{n+1}(X, A; M)$ denote the connecting homomorphism in relative cohomology. Then the following diagram is commutative:

$$\begin{tikzcd}
H^n(A;M) \arrow[r, "\delta"] \arrow[d, "h"'] & H^{n+1}(X,A;M) \arrow[d, "h"] \\
\mathrm{Hom}_R(H_n(A),M) \arrow[r, "\partial^*"'] & \mathrm{Hom}_R(H_{n+1}(X,A);M)
\end{tikzcd}$$

where h is the canonical homomorphism given by the universal coefficient theorem.

7 Variants of Cohomology

7.1 Cellular Cohomology

Definition 7.1.1 (Cellular Cochain Complex) Let X be a CW complex. Let R be a ring. Let M be an R-module. Define the cellular cochain complex of X with coefficients in M to be

$$C^n_{\text{CW}}(X;M) = \text{Hom}_R(C^{\text{CW}}_n(X), M)$$

the dual of the cellular chain complex.

TBA: We can splice the dual of the exact sequence to obtain the cellular chain complex similar to that of cellular homology.

Definition 7.1.2 (Cellular Cohomology Groups) Let X be a CW complex. Let R be a ring. Let M be an R-module. Define the n cellular cohomology group of X with coefficients in M

$$H^n_{\text{CW}}(X;M) = H^n(\text{Hom}(C^{\text{CW}}_n(X), M))$$

to be the nth cohomology group of the dual of the cellular chain complex.

7.2 Simplicial Cohomology

Definition 7.2.1 (Simplicial Cohomology Groups)
Let X be a Δ-complex. Let R be a ring. Let M be an R-module. Define the nth simplicial cohomology group of X

$$H^n_\Delta(X;M) = H^n(\text{Hom}(\Delta_\bullet(X), M))$$

to be the nth cohomology group of the dual of the simplicial complex $\Delta_\bullet(X)$ of X.

7.3 Cohomology with Compact Support

Definition 7.3.1 (Cochain with Compact Supports) Let X be a space. Let A be an abelian group. Define the nth cochain with compact supports to be the cochain subcomplex given by

$$C^n_c(X;A) = \left\{ c \in C^n(X;A) \;\middle|\; \substack{\text{there exists } K \text{ compact} \\ \text{such that } c(x)=0 \text{ for } x \notin K} \right\} \subseteq C^n(X;A)$$

and the induced coboundary map.

Definition 7.3.2 (Cohomology with Compact Supports) Let X be a space. Let A be an abelian group. Define the nth cohomology of compact supports of X to be the $n$the cohomology

$$H^n_c(X;A) = H^n(C^n_c(X;A)) = \frac{\ker(\delta : C^n_c(X;A) \to C^{n+1}_c(X;A))}{\text{im}(\delta : C^{n-1}_c(X;A) \to C^n_c(X;A))}$$

of the cochain with compact supports.

We will need an explicit construction of colimits from category theory. Recall that a directed set consists of a set I and a partial order $\leq$ such that for any $i, j \in I$, there exists $k \in I$ such that $i \leq k$ and $j \leq k$. Such a set can be made into a small category whose hom sets are given by $\text{Hom}(i,j) = * = \{f_{ij}\}$ if $i \leq j$ and $\emptyset$ otherwise. A direct system of abelian groups is then a functor $F : I \to \mathbf{Ab}$.

The colimit of the functor is then a certain cone satisfying certain universal properties. Explicitly, we can define it as

$$\operatorname*{colim}_I F = \frac{\bigoplus_{i \in I} F(i)}{\sim}$$

where the equivalence relation is generated by $x_i \sim F(f_{ij})(x_i)$. An element of the colimit is an equivalence class $[x]$ for some $x \in F(i)$ for some i. Moreover, two elements $x \in F(i)$ and $y \in F(j)$ in the direct sum $\bigoplus_{i \in I} F(i)$ are equivalent under the colimit if there exists $k \in I$ such that $i, j \leq k$ and

$$f_{ik}(x) = f_{jk}(y)$$

Proposition 7.3.3 Let X be a space. Let A be an abelian group. Consider the directed system $I = \{K \subseteq X \mid K \text{ is compact }\}$ of subspaces of X. Then there is an canonical isomorphism

$$\operatorname*{colim}_{K \in I} H^n(X, X \setminus K; A) \cong H_c^n(X; A)$$

induced by the universal property for any $n \in \mathbb{N}$.

Proof Firstly, notice that since each $C_c^n(X, X \setminus K; A)$ is a subgroup of $C_c^n(X; A)$, we have that

$$C^n(X; A) = \bigcup_{\substack{K \subseteq X \\ K \text{ compact}}} C_c^n(X, X \setminus K; A) = \operatorname*{colim}_{K \in I} C_c^n(X, X \setminus K; A)$$

I claim that if I is a directed set, and $A, B, C : I \to \mathbf{Ab}$ are functors with natural transformation $f : A \Rightarrow B$ and $g : B \Rightarrow C$ such that

$$A(i) \overset{f(i)}{\to} B(i) \overset{g(i)}{\to} C(i)$$

is exact for all i, then

$$\operatorname*{colim}_{i \in I} A(i) \overset{f}{\to} \operatorname*{colim}_{i \in I} B(i) \overset{g}{\to} \operatorname*{colim}_{i \in I} C(i)$$

Let $[x]$ be an element of $\operatorname*{colim}_{i \in I} A(i)$. Suppose that $x \in A(i)$. Then $g(f([x])) = g([f_i(x)]) = [g_i(f_i(x))] = [0]$. Hence $\operatorname{im}(f) \subseteq \ker(g)$. Now assume that $[x] \in \ker(g)$, where $x \in F(i)$. This means that $[g_i(x)] = g([x]) = [0]$. Write $\operatorname{Hom}(i, j) = \{\eta_{ij}\}$ for the unique morphism in the category I when $i \leq j$. Then $[g_i(x)] = [0]$ means that there exists $j \in I$ such that $i \leq j$ and $C(\eta_{ij})(g_i(x)) = 0$. By commutativity of the following diagram:

$$\begin{tikzcd} B(i) \arrow[r, "g_i"] \arrow[d, "B(\eta_{ij})"'] & C(i) \arrow[d, "C(\eta_{ij})"] \\ B(j) \arrow[r, "g_j"'] & C(j) \end{tikzcd}$$

we have that $0 = C(\eta_{ij})(g_{ij}(x)) = g_j(B(\eta_{ij})(x))$. Since $\operatorname{im}(f_j) = \ker(g_j)$, there exists $y \in A(j)$ such that $f_j(y) = B(\eta_{ij}(x))$. Then we have

$$f([y]) = [f_j(y)] = [B(\eta_{ij})(x)] = [x]$$

so that $\ker(g) \subseteq \operatorname{im}(f)$. Thus we are done. The result then extends to short exact sequences of abelian groups.

For the isomorphism on cohomology, we use the fact that there are exact sequences of the form:

$$0 \longrightarrow \ker(f) \longrightarrow A \overset{f}{\longrightarrow} B \longrightarrow 0$$

$$0 \longrightarrow A \overset{f}{\longrightarrow} B \longrightarrow \text{coker}(f) \longrightarrow 0$$

$$0 \longrightarrow B \overset{\text{incl.}}{\longrightarrow} A \overset{\text{proj.}}{\longrightarrow} A/B \longrightarrow 0$$

to conclude that $\operatorname*{colim}_{i \in I}$ commutes with kernels, cokernels and quotients. And then we compute that

$$\begin{aligned}
H_c^n(X;A) &= \frac{\ker(\delta_c^n : C_c^n(X;A) \to C_c^{n+1}(X;A))}{\operatorname{im}(\delta_c^n : C_c^{n-1}(X;A) \to C_c^n(X;A))} \\
&\cong \frac{\ker\left(\operatorname{colim}_{K\in I} \delta_K : C_c^n(X, X\setminus K; A) \to C_c^{n+1}(X, X\setminus K; A)\right)}{\operatorname{im}\left(\operatorname{colim}_{K\in I} \delta_K : C_c^{n-1}(X, X\setminus K; A) \to C_c^n(X, X\setminus K; A)\right)} \\
&\cong \frac{\operatorname{colim}_{K\in I} \ker(\delta_K)}{\operatorname{colim}_{K\in I} \operatorname{im}(\delta_K)} \\
&\cong \operatorname{colim}_{K\in I} \frac{\ker(\delta_K)}{\operatorname{im}(\delta_K)} \\
&= \operatorname{colim}_{K\in I} H^n(X, X\setminus K; A)
\end{aligned}$$

■

7.4 Axioms for Unreduced Cohomology

Definition 7.4.1 (Generalized Cohomology Theory for CW Pairs) A Generalized cohomology theory is a collection of contravariant functors and natural transformations

$$h^n : \mathbf{CW}^2 \to \mathbf{Ab} \quad \text{and} \quad \delta^n : h^n \circ F \Rightarrow h^{n+1}(X, A)$$

where $F(X, A) = (A, \emptyset)$, for each $n \in \mathbb{N}$ such that the following are true.

- Homotopy Invariance: If $f \simeq g : (X, A) \to (Y, B)$ then

$$h^n(f) = h^n(g) : h^n(X, A) \to h^n(Y, B)$$

- Exactness: If X is a CW-complex and $A \subseteq X$, then there is a short exact sequence

$$\cdots \longrightarrow h^n(X, A) \longrightarrow h^n(X, \emptyset) \longrightarrow h^n(A, \emptyset) \xrightarrow{\partial_n} h^{n+1}(X, A) \longrightarrow h^{n+1}(X, \emptyset) \longrightarrow \cdots$$

- Excision: If $\overline{E} \subseteq A^\circ \subseteq X$, then

$$h^n(X \setminus E, A \setminus E) \cong h^n(X, A)$$

  induced by the inclusion map

Definition 7.4.2 (Additive Cohomology Theories) Let $h_n : \mathbf{CW}^2 \to \mathbf{Ab}$ be a generalized cohomology theory. We say that h_n is additive if it satisfies the following axiom.

- Additivity: Denote the inclusion map as $\iota_k : (X_k, A_k) \to \coprod_{k\in I}(X_k, A_k)$. Then the product of the induced maps of inclusions induce an isomorphism

$$\prod_{k\in I} (\iota_k)^* : h^n\left(\coprod_{k\in I}(X_k, A_k)\right) \xrightarrow{\cong} \prod_{k\in I} h^n(X_k, A_k)$$

8 The Ring Structure in Singular Cohomology

8.1 The Cup Product

What separates singular cohomology from singular homology is that the abelian group arising from the cohomology groups has a natural structure of a ring. This in turns makes singular cohomology an invariant that associates to every space a graded ring, instead of just a sequence of abelian groups.

For this, we define the cup product which serves as multiplication between elements of the graded abelian groups.

Definition 8.1.1 (Cup Product) Let X be a space. Let R be a ring. Let $\phi \in C^k(X;R)$ and $\psi \in C^l(X;R)$. Define the cup product to be $\phi \smile \psi : C^{k+l}(X;R) \to R$ where for $\sigma = [v_0, \dots, v_{k+l}]$, we have that

$$(\phi \smile \psi)(\sigma) = \phi(\sigma|_{[v_0,\dots,v_k]}) \cdot \psi(\sigma|_{[v_k,\dots,v_{k+l}]})$$

and then extended R-linearly.

Notice that the cup product is defined in terms of the singular complexes, therefore this definition only works for chain complexes that arise from a space. Moreover, the cup product is only defined when we take singular cohomology in a ring instead of an abelian group.

Lemma 8.1.2 Let X be a space. Let R be a ring. Let $\phi \in C^k(X;R)$ and $\psi \in C^l(X;R)$. Then we have that

$$\delta(\phi \smile \psi) = \delta\phi \smile \psi + (-1)^k \phi \smile \delta\psi$$

Proof Let $\sigma : \Delta^{k+1+l} \to X$ be a singular $(k+1+l)$-simplex. We have that

$$\begin{aligned}
(\delta\phi \smile \psi)(\sigma) &= \delta(\phi)(\sigma|_{v_0,\dots,v_{k+1}}) \cdot \psi(\sigma|_{v_{k+1},\dots,v_{k+1+l}}) \\
&= \phi(\partial(\sigma|_{v_0,\dots,v_{k+1}})) \cdot \psi(\sigma|_{v_{k+1},\dots,v_{k+1+l}}) \\
&= \phi\left(\sum_{i=0}^{k+1}(-1)^i \sigma|_{v_0,\dots,\hat{v}_i,\dots,v_{k+1}}\right) \cdot \psi(\sigma|_{v_{k+1},\dots,v_{k+1+l}}) \\
&= \sum_{i=0}^{k+1}(-1)^i \phi(\sigma|_{v_0,\dots,\hat{v}_i,\dots,v_{k+1}}) \cdot \psi(\sigma|_{v_{k+1},\dots,v_{k+1+l}})
\end{aligned}$$

and

$$\begin{aligned}
(-1)^k\left(\phi \smile \delta(\psi)\right)(\sigma) &= \phi(\sigma|_{v_0,\dots,v_k}) \cdot (\psi(\partial\sigma|_{v_k,\dots,v_{k+1+l}})) \\
&= \phi(\sigma|_{v_0,\dots,v_k}) \cdot ((-1)^k \psi(\partial\sigma|_{v_k,\dots,v_{k+1+l}})) \\
&= \phi(\sigma|_{v_0,\dots,v_k}) \cdot \psi\left(\sum_{j=k}^{k+1+l}(-1)^j \sigma|_{v_k,\dots,\hat{v}_j,\dots,v_{k+1+l}}\right) \\
&= \phi(\sigma|_{v_0,\dots,v_k}) \cdot \sum_{j=k}^{k+1+l}(-1)^j \psi(\sigma|_{v_k,\dots,\hat{v}_j,\dots,v_{k+1+l}})
\end{aligned}$$

Notice that the last term of the first sum cancels out the first term of the second sum. Thus we have

$$\begin{aligned}
&(\delta\phi \smile \psi)(\sigma) + (-1)^k\left(\phi \smile \delta(\psi)\right)(\sigma) \\
&= \sum_{i=0}^{k+1}(-1)^i \phi(\sigma|_{v_0,\dots,\hat{v}_i,\dots,v_{k+1}}) \cdot \psi(\sigma|_{v_{k+1},\dots,v_{k+1+l}}) + \phi(\sigma|_{v_0,\dots,v_k}) \cdot \sum_{j=k}^{k+1+l}(-1)^j \psi(\sigma|_{v_k,\dots,\hat{v}_j,\dots,v_{k+1+l}}) \\
&= \sum_{i=0}^{k}(-1)^i \phi(\sigma|_{v_0,\dots,\hat{v}_i,\dots,v_{k+1}}) \cdot \psi(\sigma|_{v_{k+1},\dots,v_{k+1+l}}) + \phi(\sigma|_{v_0,\dots,v_k}) \cdot \sum_{j=k+1}^{k+1+l}(-1)^j \psi(\sigma|_{v_k,\dots,\hat{v}_j,\dots,v_{k+1+l}}) \\
&= (\phi \smile \psi)(\partial\sigma) \\
&= \delta(\phi \smile \psi)(\sigma)
\end{aligned}$$

and so we conclude. ■

Proposition 8.1.3 Let X be a space. Let R be a ring. Then the cup product

$$\smile: H^m(X;R) \times H^n(X;R) \to H^{m+n}(X;R)$$

descends to homology and is well defined.

Proof By the universal property of quotients, we just have to show the following:

- If ϕ and ψ are cochains, then $\phi \smile \psi$ is a cochain
- If either ϕ or ψ are coboundaries, then $\phi \smile \psi$ is a coboundary.

So let ϕ and ψ are cochains. Then $\delta(\phi) = 0$ and $\delta(\psi) = 0$. By the above lemma, we have that

$$\delta(\phi \smile \psi) = 0 \smile \psi + (-1)^n \phi \smile 0 = 0$$

Hence we are done.

If ϕ is a coboundary, then there exists $\alpha \in C^{n-1}(X;R)$ such that $\delta(\alpha) = \phi$. Then we have that

$$\delta(\alpha \smile \psi) = \phi \smile \psi + (-1)^{n-1}\alpha \smile 0 = \psi \smile \psi$$

so that $\phi \smile \psi$ is a coboundary. If ψ is a coboundary, then there exists $\beta \in C^{m-1}(X;R)$ such that $\delta(\beta) = \psi$. Then we have that

$$\delta(\phi \smile \beta) = 0 \smile \beta + (-1)^n \phi \smile \psi$$

so that $\phi \smile \psi$ is a coboundary. ■

Proposition 8.1.4 Let X be a space and let R be a commutative ring. If $[\phi] \in H^m(X;R)$ and $[\psi] \in H^n(X;R)$, then

$$[\phi] \smile [\psi] = (-1)^{mn}[\psi] \smile [\phi]$$

Proof Let $\varepsilon_n = (-1)^{\frac{n(n+1)}{2}}$. Let L be the linear homeomorphism reversing the order of the vertices of $[v_0, \dots, v_n]$. It sends the vertex v_i to v_{n-i}. Define $\rho : C_n(X) \to C_n(X)$ by

$$\rho(\sigma) = \varepsilon_n(\sigma \circ L)$$

We first show that ρ is a chain map. On one hand, we have

$$\partial(\rho(\sigma)) = \varepsilon_n \sum_{k=0}^{n} (-1)^k (\sigma \circ L)|_{[v_0,\dots,\widehat{v}_k,\dots,v_n]}$$

and

$$\begin{aligned}
\rho(\partial\sigma)) &= \rho\left(\sum_{k=0}^{n}(-1)^k\sigma|_{[v_0,\dots,\widehat{v}_k,\dots,v_n]}\right)\\
&= \sum_{k=0}^{n}(-1)^k\varepsilon_{n-1}(\sigma|_{[v_0,\dots,\widehat{v}_k,\dots,v_n]}\circ L)\\
&= \varepsilon_{n-1}\sum_{k=0}^{n}(-1)^k(\sigma\circ L)|_{[v_0,\dots,\widehat{v}_{n-k},\dots,v_n]}\\
&= \varepsilon_{n-1}\sum_{k=0}^{n}(-1)^{n-k}(\sigma\circ L)|_{[v_0,\dots,\widehat{v}_k,\dots,v_n]}\\
&= (-1)^n\varepsilon_{n-1}\sum_{k=0}^{n}(-1)^k(\sigma\circ L)|_{[v_0,\dots,\widehat{v}_k,\dots,v_n]}\\
&= \varepsilon_n\sum_{k=0}^{n}(-1)^k(\sigma\circ L)|_{[v_0,\dots,\widehat{v}_k,\dots,v_n]}\\
&= \partial(\rho(\sigma))
\end{aligned}$$

so that ρ is a chain map. We now want to show that ρ is chain homotopic to $\mathrm{id}: C_n(X)\to C_n(X)$. To show this, define $P: C_n(X)\to C_{n+1}(X)$ by the formula

$$P(\sigma|_{[v_0,\dots,v_n]}) = \sum_{k=0}^{n}(-1)^k\varepsilon_{n-k}(\sigma\circ\pi)|_{[v_{00},\dots,v_{k0},v_{n1},\dots,v_{k1}]}$$

where $\pi:\Delta^n\times I\to\Delta^n$ is the projection map. We compute that

$$\begin{aligned}
\partial(P(\sigma|_{[v_0,\dots,v_n]})) &= \partial\left(\sum_{k=0}^{n}(-1)^k\varepsilon_{n-k}(\sigma\circ\pi)|_{[v_{00},\dots,v_{k0},v_{n1},\dots,v_{k1}]}\right)\\
&= \sum_{k=0}^{n}(-1)^k\varepsilon_{n-k}\partial((\sigma\circ\pi)|_{[v_{00},\dots,v_{k0},v_{n1},\dots,v_{k1}]})\\
&= \sum_{k=0}^{n}(-1)^k\varepsilon_{n-k}\Big(\sum_{j=0}^{k}(-1)^j(\sigma\circ\pi)|_{[v_{00},\dots,\widehat{v}_{j0},\dots,v_{k0},v_{n1},\dots,v_{k1}]}\\
&\quad+\sum_{j=k+1}^{n+1}(-1)^j(\sigma\circ\pi)|_{[v_{00},\dots,v_{k0},v_{n1},\dots,\widehat{v}_{n-j+1+k,1},\dots,v_{k1}]}\Big)\\
&= \sum_{k=0}^{n}(-1)^k\varepsilon_{n-k}\Big(\sum_{j=0}^{k}(-1)^j(\sigma\circ\pi)|_{[v_{00},\dots,\widehat{v}_{j0},\dots,v_{k0},v_{n1},\dots,v_{k1}]}\\
&\quad+\sum_{j=k}^{n}(-1)^{n-j+1+k}(\sigma\circ\pi)|_{[v_{00},\dots,v_{k0},v_{n1},\dots,\widehat{v}_{j1},\dots,v_{k1}]}\Big)
\end{aligned}$$

The term for which $k = j$ is given by

$$\begin{aligned}
&\varepsilon_n(\sigma\circ\pi)|_{[v_{n1},\dots,v_{k1}]} + \sum_{k=1}^{n}(-1)^k\varepsilon_{n-k}(-1)^k(\sigma\circ\pi)|_{[v_{00},\dots,v_{k-1,0},v_{n1},\dots,v_k]} \\
&\quad + \sum_{k=0}^{n-1}(-1)^k(-1)^{n-1}\varepsilon_{n-k}(\sigma\circ\pi)|_{[v_{00},\dots,v_{k0},v_{n1},\dots,v_{k+1,1}]} - [v_{00},\dots,v_{n0}] \\
&= \varepsilon_n[v_{n1},\dots,v_{k1}] + \sum_{k=1}^{n}\varepsilon_{n-k}(\sigma\circ\pi)|_{[v_{00},\dots,v_{k-1,0},v_{n1},\dots,v_k]} \\
&\quad + \sum_{k=0}^{n-1}(-1)^{n+k-1}\varepsilon_{n-k}(\sigma\circ\pi)|_{[v_{00},\dots,v_{k0},v_{n1},\dots,v_{k+1,1}]} - [v_{00},\dots,v_{n0}]
\end{aligned}$$

Notice that in both sums, the coefficients of $(\sigma\circ\pi)|_{[v_{00},\dots,v_{k0},v_{n1},\dots,v_{k+1,1}]}$ are ε_{n-k-1} and $(-1)^{n+k-1}\varepsilon_{n-k}$. Their sum is 0 so these terms in the double sum cancel out. We are left with $\partial(P(\sigma|_{[v_0,\dots,v_n]}))$ equal to

$$\begin{aligned}
&\sum_{k=0}^{n}\varepsilon_{n-k}\left(\sum_{j=0}^{k-1}(-1)^{k+j}(\sigma\circ\pi)|_{[v_{00},\dots,\widehat{v}_{j0},\dots,v_{k0},v_{n1},\dots,v_{k1}]} + \sum_{j=k+1}^{n}(-1)^{n-j+1}(\sigma\circ\pi)|_{[v_{00},\dots,v_{k0},v_{n1},\dots,\widehat{v}_{j1},\dots,v_{k1}]}\right) \\
&\quad + \varepsilon_n(\sigma\circ\pi)|_{[v_{n1},\dots,v_{01}]} - (\sigma\circ\pi)|_{[v_0,\dots,v_n]}
\end{aligned}$$

One can compute $P(\partial(\sigma|_{[v_0,\dots,v_n]})$ to get

$$\partial(P(\sigma|_{[v_0,\dots,v_n]} + P(\partial(\sigma|_{[v_0,\dots,v_n]})) = \varepsilon_n\sigma|_{[v_n,\dots,v_0]} - \sigma|_{[v_0,\dots,v_n]} = \rho(\sigma|_{[v_0,\dots,v_n]}) - \mathrm{id}(\sigma|_{[v_0,\dots,v_n]})$$

Let $\phi\in C^k(X;R)$ and $\psi\in C^l(X;R)$. Then we have that

$$\begin{aligned}
(\rho^*(\phi)\smile\rho^*(\psi))(\sigma) &= \phi(\varepsilon_k\sigma|_{[v_k,\dots,v_0]})\psi(\varepsilon_l\sigma|_{[v_{k+l},\dots,v_k]}) \\
&= \varepsilon_k\varepsilon_l\phi(\sigma|_{[v_k,\dots,v_0]})\psi(\sigma|_{[v_{k+l},\dots,v_k]})
\end{aligned}$$

Since $\varepsilon_k\varepsilon_l = (-1)^{kl}\varepsilon_{k+l}$ and R is assumed commutative, we have that

$$\begin{aligned}
\varepsilon_k\varepsilon_l\phi(\sigma|_{[v_k,\dots,v_0]})\psi(\sigma|_{[v_{k+l},\dots,v_k]}) &= (-1)^{kl}\varepsilon_{k+l}\psi(\sigma|_{[v_{k+l},\dots,v_k]})\phi(\sigma|_{[v_k,\dots,v_0]}) \\
&= (-1)^{kl}\rho^*(\psi\smile\phi)(\sigma)
\end{aligned}$$

Since ρ^* and id are chain homotopic, they descend to the identity

$$[\phi]\smile[\psi] = (-1)^{kl}[\psi]\smile[\phi]$$

■

It is important to note that the above proposition (graded commutativity) is only true when R is a commutative ring.

Theorem 8.1.5 (The Cohomology Ring) Let X be a topological space and R a commutative ring with identity. Then the direct sum of abelian groups

$$H^*(X;R) = \bigoplus_{i=0}^{\infty}H^i(X;R)$$

is a graded commutative ring with identity $1_R\in H^0(X;R)$. The product is given by the cup product

$$\smile\colon H^m(X;R)\times H^n(X;R)\to H^{m+n}(X;R)$$

defined by
$$(\phi \smile \psi)(\sigma) = \phi(\sigma|_{[v_0,\dots,v_n]}) \cdot \psi(\sigma|_{[v_n,\dots,v_{n+m}]})$$
Moreover, $H^*(X;R)$ is an R-algebra.

Proof

Lemma 8.1.6 Let X,Y be spaces. Let R be a commutative ring. Let $f : X \to Y$ be a map. Then the induced map
$$f^* : H^*(Y;R) \to H^*(X;R)$$
is a graded ring homomorphism.

Example 8.1.7 Let $n \in \mathbb{N}$. Then there is an isomorphism of graded rings
$$H^*(S^n;\mathbb{Z}) \cong \frac{\mathbb{Z}[t]}{(t^2)}$$
where t is of degree n in the graded structure.

Example 8.1.8 Let $n \geq 1$. Then the following are true.

- There is an isomorphism of graded rings
$$H^*(\mathbb{RP}^n;\mathbb{Z}/2\mathbb{Z}) \cong \frac{\mathbb{Z}/2\mathbb{Z}[t]}{(t^{n+1})}$$
where t is of degree 1 in the graded structure.
- There is an isomorphism of graded rings
$$H^*(\mathbb{CP}^n;\mathbb{Z}) \cong \frac{\mathbb{Z}[t]}{(t^{n+1})}$$
where t is of degree 2 in the graded structure.

Proposition 8.1.9 Let $\mathbb{R}^n$ be equipped with the structure of a division algebra over $\mathbb{R}$. Then n is a power of 2.

Proposition 8.1.10
Let X and Y be spaces. Then we have an isomorphism of graded rings
$$H^*(X \vee Y) \cong \frac{H^*(X) \times H^*(Y)}{\langle 1_X, 1_Y \rangle}$$
where 1_X and 1_Y are the identities of $H^*(X)$ and $H^*(Y)$ respectively.

Example 8.1.11
$\mathbb{CP}^2$ and $S^2 \vee S^4$ are not homotopy equivalent.

Proof
We note that $H^*(S^2 \vee S^4) \cong \frac{\mathbb{Z}[a,b]}{(a^2,b^2,ab)}$. This ring is not isomorphic to $H^*(\mathbb{CP}^2) \cong \frac{\mathbb{Z}[t]}{(t^3)}$.

8.2 The Relative Cup Product

Definition 8.2.1 (The Relative Cup Product) Let X be a space. Let $A, B \subseteq X$ be open. Let R be a ring. Define the relative cup product of A and B to be the map
$$\smile : H^k(X,A;R) \times H^l(X,B;R) \to H^{k+l}(X, A \cup B;R)$$

by the formula

$$(\phi \smile \psi)(\sigma) = \phi(\sigma|_{[v_0,\dots,v_k]}) \cdot \psi(\sigma|_{[v_k,\dots,v_{k+l}]})$$

for $\phi \in C^k_{\text{Sing}}(X, A; R)$ and $\psi \in C^l_{\text{Sing}}(X, B; R)$.

In particular, notice that this is the same formula for the (absolute) cup product, other than the fact that the domain of the two inputs are restricted. But we also want to show that this map is well defined in the sense that the codomain is given correctly.

Lemma 8.2.2 Let X be a space. Let $A, B \in X$ be open. Let R be a ring. Then the relative cup product

$$\smile : H^k(X, A; R) \times H^l(X, B; R) \to H^{k+l}(X, A \cup B; R)$$

is well defined.

Proof Let $\phi \in C^k_{\text{Sing}}(X, A; R)$ and $\psi \in C^l_{\text{Sing}}(X, B; R)$. Then ϕ vanishes on any simplexes in A. Likewise ψ vanishes on any simplexes in B. Recall in the proof of excision that $C^n(X, A+B; R)$ is the subgroup of $C^n(X; R)$ consisting of cochains that vanish on sums of chains in A and sums of chains in B. We conclude that $\phi \smile \psi \in C^{k+l}(X, A+B; R)$. So now our cup product is well-defined in the following way:

$$\smile : C^k(X, A; R) \times C^l(X, B; R) \to C^{k+l}(X, A+B; R)$$

Moreover, they descend to their respective cohomology groups in the same way as 11.4.3.

Recall from singular homology that the following two horizontal sequences are exact:

$$0 \longrightarrow C_n(A \cup B) \hookrightarrow C_n(X) \twoheadrightarrow C_n(X, A \cup B) \longrightarrow 0$$

$$0 \longrightarrow C_n(A + B) \hookrightarrow C_n(X) \twoheadrightarrow C_n(X, A + B) \longrightarrow 0$$

Dualizing also gives two short exact sequence of chain complexes:

$$0 \longrightarrow C^n(X, A \cup B; R) \hookrightarrow C^n(X; R) \twoheadrightarrow C^n(A \cup B; R) \longrightarrow 0$$

$$0 \longrightarrow C^n(X, A + B; R) \hookrightarrow C^n(X; R) \twoheadrightarrow C^n(A + B; R) \longrightarrow 0$$

Now recall from 4.3.7 that the inclusion $C_\bullet(A+B) \to C_\bullet(A \cup B)$ is chain homotopic to the identity map. Dualizing the inclusion gives a restriction map $C^\bullet(A \cup B) \to C^\bullet(A+B)$ that is chain homotopic to the identity. By naturality of the connecting homomorphism, we now obtain the following diagram:

$$\begin{tikzcd}
\cdots \arrow[r] & H^n(X, A\cup B; R) \arrow[r] \arrow[d, "\cong"'] & H^n(X; R) \arrow[r] \arrow[d, "\cong"] & H^n(A \cup B; R) \arrow[r, "\delta"] \arrow[d] & H^{n+1}(X, A \cup B; R) \arrow[r] \arrow[d, "\cong"'] & H^{n+1}(X; R) \arrow[r] \arrow[d, "\cong"] & \cdots \\
\cdots \arrow[r] & H^n(X, A+B; R) \arrow[r] & H^n(X; R) \arrow[r] & H^n(A+B; R) \arrow[r, "\delta"] & H^{n+1}(X, A+B; R) \arrow[r] & H^{n+1}(X; R) \arrow[r] & \cdots
\end{tikzcd}$$

By the 5-lemma, we conclude that the middle map is an isomorphism and so we conclude. ∎

Theorem 8.2.3 (The Relative Cohomology Ring) Let X be a topological space and R a commutative ring with identity. Then the direct sum of abelian groups

$$H^*(X; R) = \bigoplus_{i=0}^{\infty} H^i(X; R)$$

is a graded commutative ring with identity $1_R \in H^0(X; R)$. The product is given by the cup product

$$\smile : H^m(X; R) \times H^n(X; R) \to H^{m+n}(X; R)$$

defined by

$$(\phi \smile \psi)(\sigma) = \phi(\sigma|_{[v_0,\dots,v_n]}) \cdot \psi(\sigma|_{[v_n,\dots,v_{n+m}]})$$

Moreover, $H^*(X; R)$ is an R-algebra.

Proof

8.3 The Kunneth Formula

Definition 8.3.1 (The Cross Product) Let X, Y be topological spaces. Let R be a commutative ring. Denote $p_1 : X \times Y \to X$ and $p_2 : X \times Y \to Y$ the projection maps. Define the cross product of $x \in H^m(X; R)$ and $y \in H^m(Y; R)$ for a ring R to be

$$x \times y = p_1^*(x) \smile p_2^*(y)$$

where $x \times y \in H^{m+n}(X \times Y; R)$.

Proposition 8.3.2 Let X, Y be topological spaces. Let R be a commutative ring. Let $a, c \in H^*(X; R)$ and $b, d \in H^*(Y; R)$ with $c \in H^m(X; R)$ and $d \in H^n(Y; R)$ we have

$$(a \times b) \smile (c \times d) = (-1)^{mn}(a \smile c) \times (b \smile d)$$

Let R be a commutative ring. Let M and N be graded R-modules. Recall that their tensor product can also be given a grading where

$$(M \otimes N)_n = \bigoplus_{i+j=n} M_i \otimes_R N_j$$

Theorem 8.3.3 (The Kunneth Formula) Let X and Y be CW-complexes. Let R be a commutative ring. Then the cross product

$$\times : H^*(X; R) \otimes_R H^*(Y; R) \to H^*(X \times Y; R)$$

defined by $a \otimes b \mapsto a \times b$ is graded ring homomorphism. Moreover, it is an isomorphism of graded rings if $H^k(Y; R)$ is a finitely generated free R-module for all k.

Example 8.3.4 There is an isomorphism of graded rings

$$H^*(T; \mathbb{Z}) \cong \frac{\mathbb{Z}\langle s, t\rangle}{(s^2, t^2, st + ts)}$$

where s, t are elements of degree 1.

Example 8.3.5
Let R be a commutative ring. Let $p, q \in \mathbb{N}^+$. There is an isomorphism of graded rings

$$H^*(S^p \times S^q; R) \cong \Lambda(Rx \oplus Ry) = \frac{R\langle x, y\rangle}{(x^2, y^2, xy + (-1)^{pq+1}yx)}$$

where $\deg(x) = p$ and $\deg(y) = q$.

Theorem 8.3.6 (The Relative Kunneth Formula) Let (X, A) and (Y, B) be CW complex pairs. Let R be a commutative ring. Then the cross product

$$\times : H^*(X, A; R) \otimes_R H^*(Y, B; R) \to H^*(X \times Y, X \times B \cup A \times Y; R)$$

is graded ring homomorphism. Moreover, it is an isomorphism of graded rings if $H^k(Y, B; R)$ is a finitely generated free R-module for all k.

8.4 Lusternik-Schnirelman Category

Definition 8.4.1 (Lusternik-Schnirelman Category) Let X be a space. Define the Lusternik-Schnirelman category of X to be the smallest $n \in \mathbb{N}$ such that there exists an open cover $X = \bigcup_{i=1}^n U_i$ so that $U_i \hookrightarrow X$ is null-homotopic.

Proposition 8.4.2 Let X be a space. Suppose that X has Lusternik-Schnirelman category $n \in \mathbb{N}$. Let $x_i \in H^{k_i}(X;\mathbb{Z})$ for $k_i > 0$. Then we have

$$x_1 \smile \cdots \smile x_n = 0$$

Proof For each U_k in the open cover, notice that $U_k \hookrightarrow X$ begin null-homotopic implies that the long exact sequence in relative cohomology becomes

$$\cdots \longrightarrow H^{i_j}(X, U_k) \longrightarrow H^{i_j}(X) \xrightarrow{0} H^{i_j}(U_k) \longrightarrow \cdots$$

which also implies that $H^{i_j}(X, U_k) \to H^{i_j}(X)$ is surjective. Thus by taking the product, we have that the map

$$H^{i_1}(X, U_1) \times \cdots \times H^{i_n}(X, U_n) \to H^{i_1}(X) \times \cdots \times H^{i_n}(X)$$

is surjective. Now by surjectivity, there exists a lift $(\tilde{x}_1, \dots, \tilde{x}_n) \in H^{i_1}(X, U_1) \times \cdots \times H^{i_n}(X, U_n)$ of $(x_1, \dots, x_n)$. Now using the relative version of the cup product, we obtain a commutative diagram:

$$\begin{tikzcd}
\prod_{k=1}^n H^{i_k}(X;R) \arrow[r, "\smile"] & H^{i_1+\cdots+i_n}(X;R) \\
\prod_{k=1}^n H^{i_k}(X,U_k;R) \arrow[r, "\smile"] \arrow[u, two heads] & H^{i_1+\cdots+i_n}(X,U_1\cup\cdots\cup U_n;R) \arrow[u, two heads]
\end{tikzcd}$$

(why is it natural / commutative) But $X = \bigcup_{i=1}^n U_i$ implies that $H^{i_1+\cdots+i_n}(X, U_1 \cup \cdots \cup U_n; R) = 0$. Thus $(x_1, \dots, x_n)$ is mapped to the 0 element. ■

9 The Euler Characteristic

The Euler characteristic is similar to a notion of size. In set theory we have cardinality, in linear algebra we have dimensions, in abelian groups we have rank. The Euler characteristic acts similar to these quantities.

9.1 The Characteristic as an Invariant

Definition 9.1.1 (The Euler Characteristic) Let X be a space with only finitely many non-zero homology groups, all of which are finitely generated abelian groups. Then the Euler characteristic is defined as

$$\chi(X)=\sum_{n\geq 0}(-1)^n\operatorname{rank}(H_n(X))$$

Example 9.1.2
The following are true.
- $\chi(S^n)=1+(-1)^n$.
- $\chi(\mathbb{RP}^n)=0$ if n is odd and $\chi(\mathbb{RP}^n)=1$ if n is even.
- $\chi(\mathbb{CP}^n)=n+1$.
- $\chi(T)=0$.

Lemma 9.1.3 Let $C_\bullet$ be a chain complex with only finitely many non-zero terms, all of which are finitely generated abelian groups. Then

$$\sum_{n\in\mathbb{Z}}(-1)^n\operatorname{rank}(C_n)=\sum_{n\in\mathbb{Z}}(-1)^n\operatorname{rank}(H_n(C_\bullet))$$

Proof We have two short exact sequences for all n given by

$$0\longrightarrow Z_n\longrightarrow C_n\longrightarrow B_{n-1}\longrightarrow 0$$

$$0\longrightarrow B_n\longrightarrow Z_n\longrightarrow H_n\longrightarrow 0$$

We therefore have that

$$\begin{aligned}\sum_{n\in\mathbb{Z}}(-1)^n\operatorname{rank}(C_n)&=\sum_{n\in\mathbb{Z}}(-1)^n(\operatorname{rank}(Z_n)+\operatorname{rank}(B_{n-1}))\\&=\sum_{n\in\mathbb{Z}}(-1)^n(\operatorname{rank}(Z_n)-\operatorname{rank}(B_n))\\&=\sum_{n\in\mathbb{Z}}(-1)^n\operatorname{rank}(H_n)\end{aligned}$$

and so we conclude. ■

Corollary 9.1.4 Let X be a CW complex. Then the alternating sum

$$\chi(X)=\sum_{n\geq 0}(-1)^n|\{n\text{-cells}\}|$$

In particular, $\chi(X)$ is independent of the choice of CW structure.

Proof By the above corollary, we have that the sum is equal to $\sum_{n\in\mathbb{Z}}(-1)^n\operatorname{rank}(H_n(X))$ and the rank of $H_n(X)$ is independent of the cell structure. ■

Lemma 9.1.5 Let X be a space with only finitely many non-zero homology groups, all of which are finitely generated abelian groups. Let $\mathbb{F}$ be a field. Then we have

$$\chi(X) = \sum_{n\geq 0}(-1)^n \dim_{\mathbb{F}}(H^n(X;\mathbb{F}))$$

Definition 9.1.6 (Plane Graph) A plane graph is a finite 1-dimensional CW-complex embedded in the real plane $\mathbb{R}^2$. Equivalently, it is a finite graph in the plane in which the edges do not cross. A planar graph is a finite 1-dimensional CW-complex that exhibits such an embedding.

A face of a graph X is a connected component of $\mathbb{R}^2 \setminus X$.

The Euler characteristic is in fact a generalization of Euler's formula.

Proposition 9.1.7 (Euler's Formula) Let X be a planar graph with v vertices, e edges and f faces. Then we have

$$v - e + f = 2$$

9.2 First Properties of the Euler Characteristic

Proposition 9.2.1 Let $X = U \cup V$ be a space and assume that one of the following holds.

- X is a CW-complex and U, V are subcomplexes
- $U, V \subseteq X$ are open

Then if $\chi(U), \chi(V), \chi(U \cap V)$ are well defined, $\chi(X)$ is also well defined and we have

$$\chi(X) = \chi(U) + \chi(V) - \chi(U \cap V)$$

Proof If U and V are finite CW complexes then so is X. Looking at the number c_n of cells in each dimension n we have that

$$c_n(X) = c_n(U) + c_n(V) - c_n(U \cap V)$$

which completes the proof.

If U and V are open, then using Mayer-Vietoris sequence, we obtain an alternating sum of cells in X, U and V. By corollary 11.1.3, it follows that since the sequence is exact, we have that

$$\sum_{n\in\mathbb{Z}}(-1)^n(\text{rank}(H_n(U \cap V)) - \text{rank}(H_n(U)) - \text{rank}(H_n(V)) + \text{rank}(H_n(X))) = 0$$

and so we conclude. ∎

Proposition 9.2.2 Let X and Y be finite CW-complexes. Then so is $X \times Y$ and we have

$$\chi(X \times Y) = \chi(X) \times \chi(Y)$$

Proof We first prove that $c_n(X \times Y) = \sum_{a+b=n} c_a(X)c_b(Y)$. The result then follows by proposition 11.1.3. ∎

We can relate the Euler characteristic with covering spaces by the following formula.

Proposition 9.2.3 Let $p : \tilde{X} \to X$ be a d-sheeted cover and X is a finite CW complex. Then we have that

$$\chi(\tilde{X}) = d \cdot \chi(X)$$

Proof In the proof that $\tilde{X}$ is also a CW complex, we have seen that if X has k amount of n-cells, then $\tilde{X}$ has $d \cdot k$ amount. This applies to all n and so we obtain the formula. ■